IMPERIAL

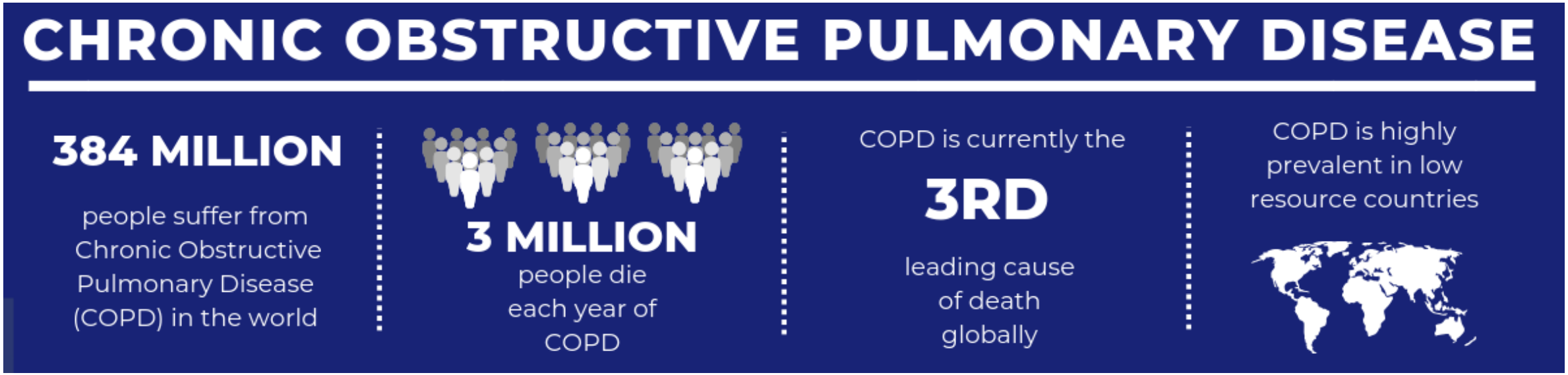
Chronic obstructive pulmonary disease

Diagnosis, Management and Respiratory Infections

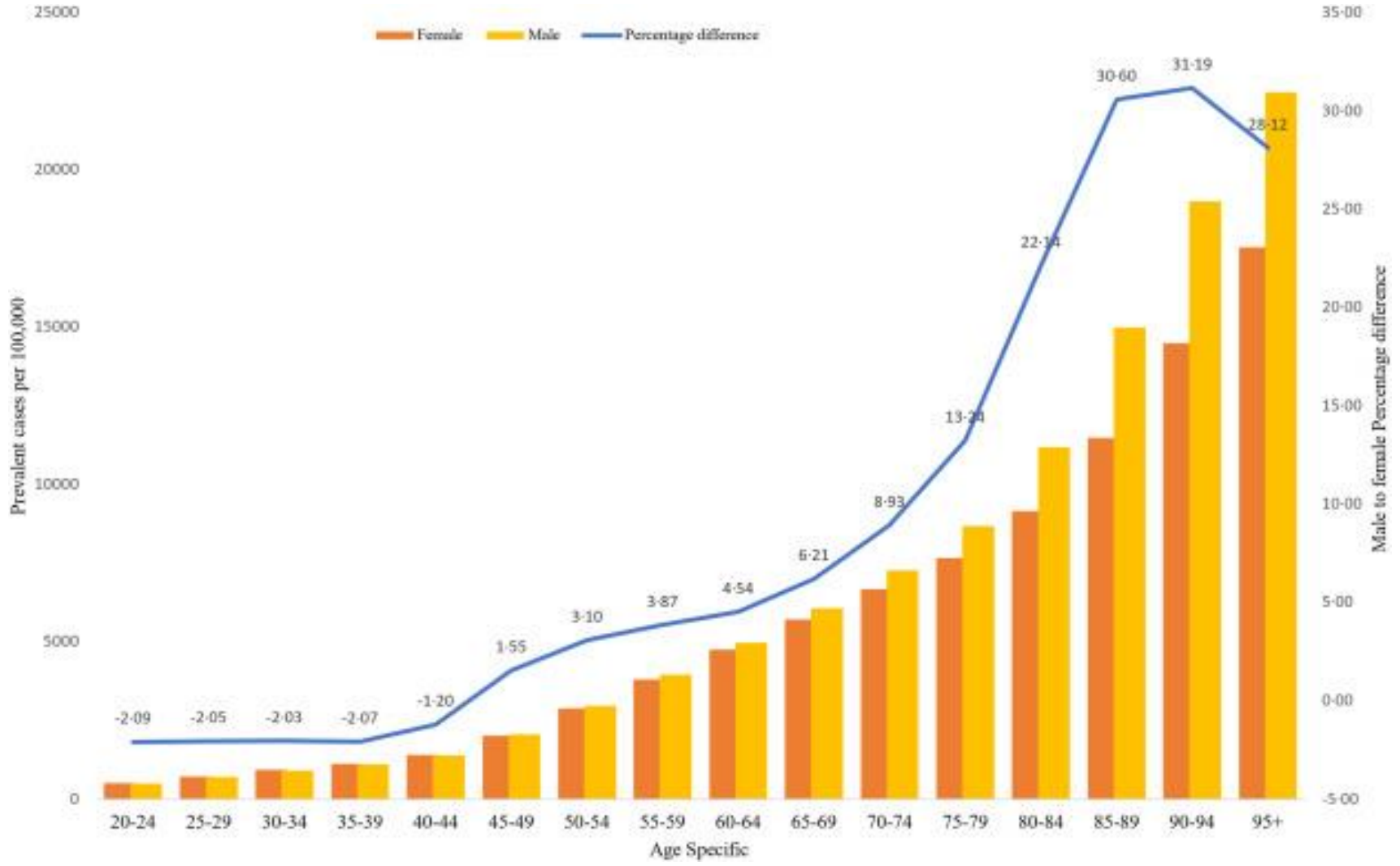
Dr Lydia Finney

4 May, 2026

COPD: A global problem



Increasing burden of COPD in Sub-Saharan Africa



Aleymayohu *et al* eClinical Medicine 2023

Chronic Obstructive Pulmonary Disease - Definition

Global initiative for Obstructive Lung Disease (GOLD) 2026

- Heterogeneous Disease
- Causes Chronic Respiratory Symptoms
- Due to abnormalities of the airways (bronchitis/ bronchiolitis) and/or alveoli (emphysema)
- Causing persistent, often progressive airflow obstruction

COPD – who is at risk?



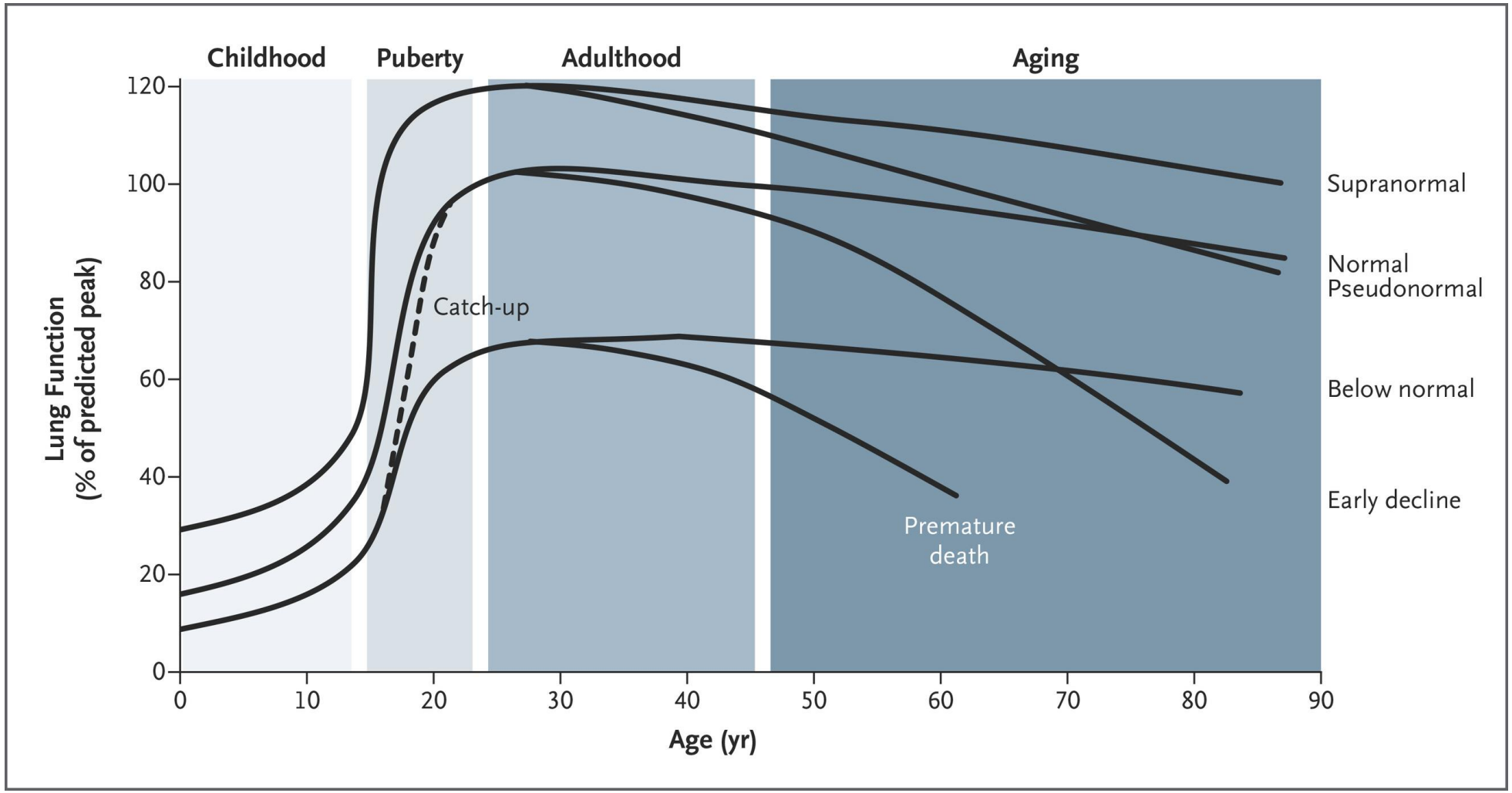








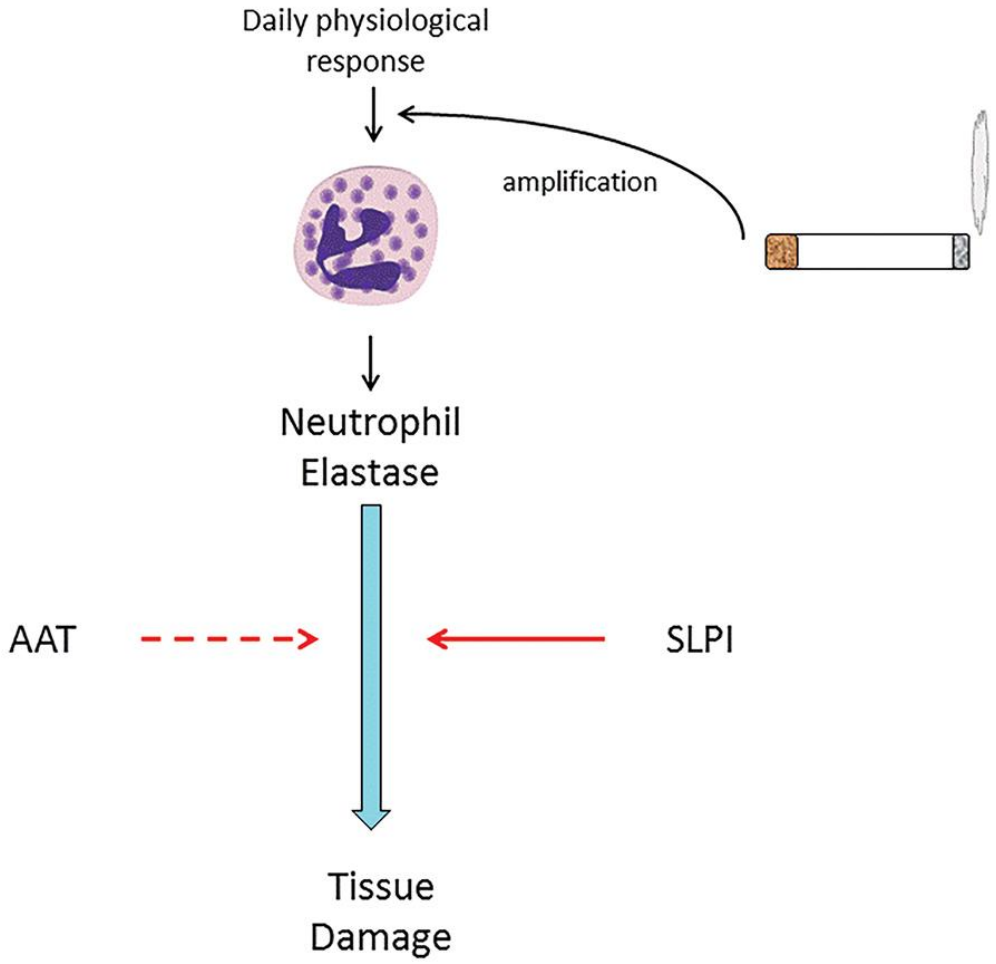
Lung function trajectories

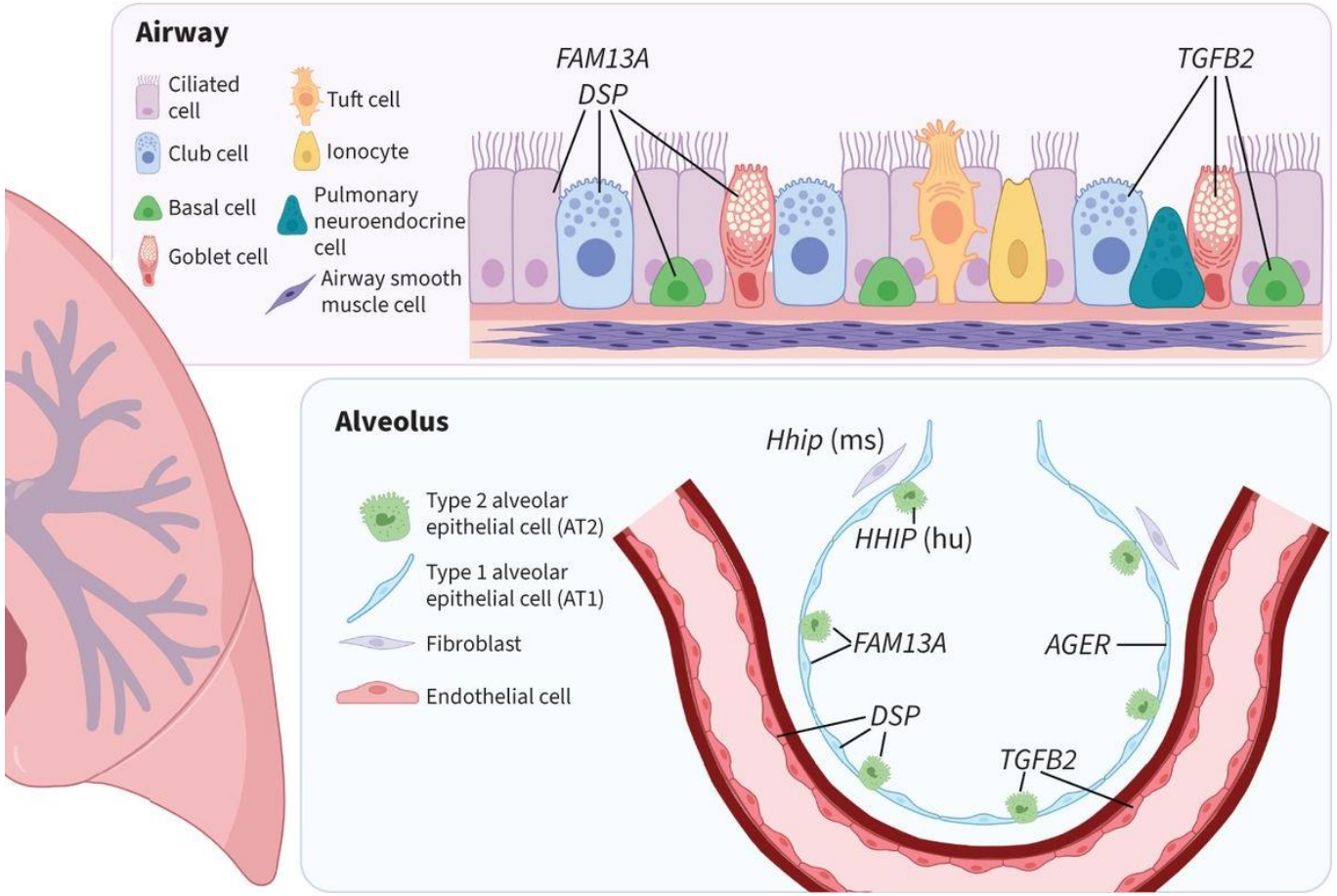


Augusti & Hogg NEJM 2019

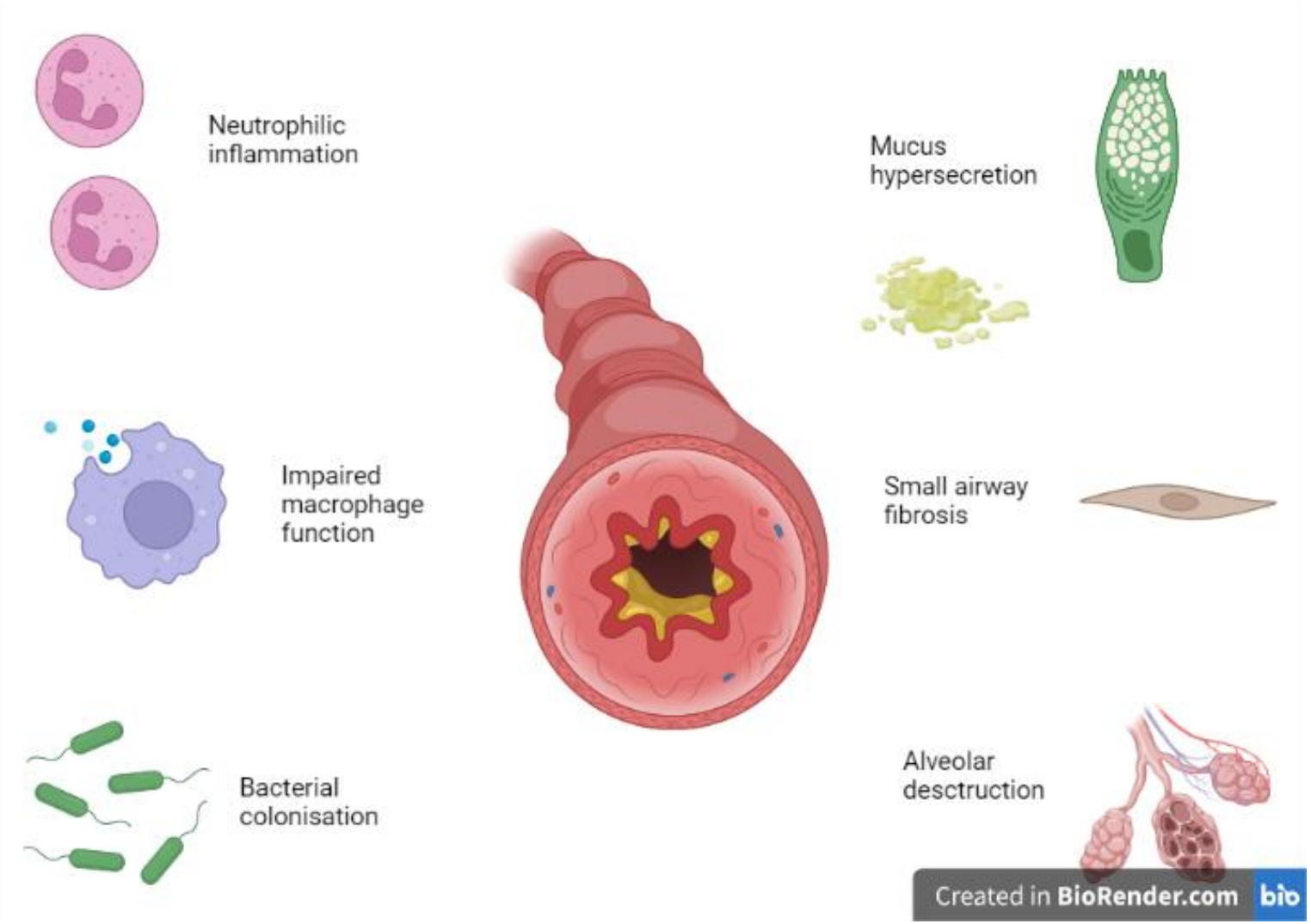
Genetics of COPD

Alpha-1 anti-trypsin deficiency and beyond



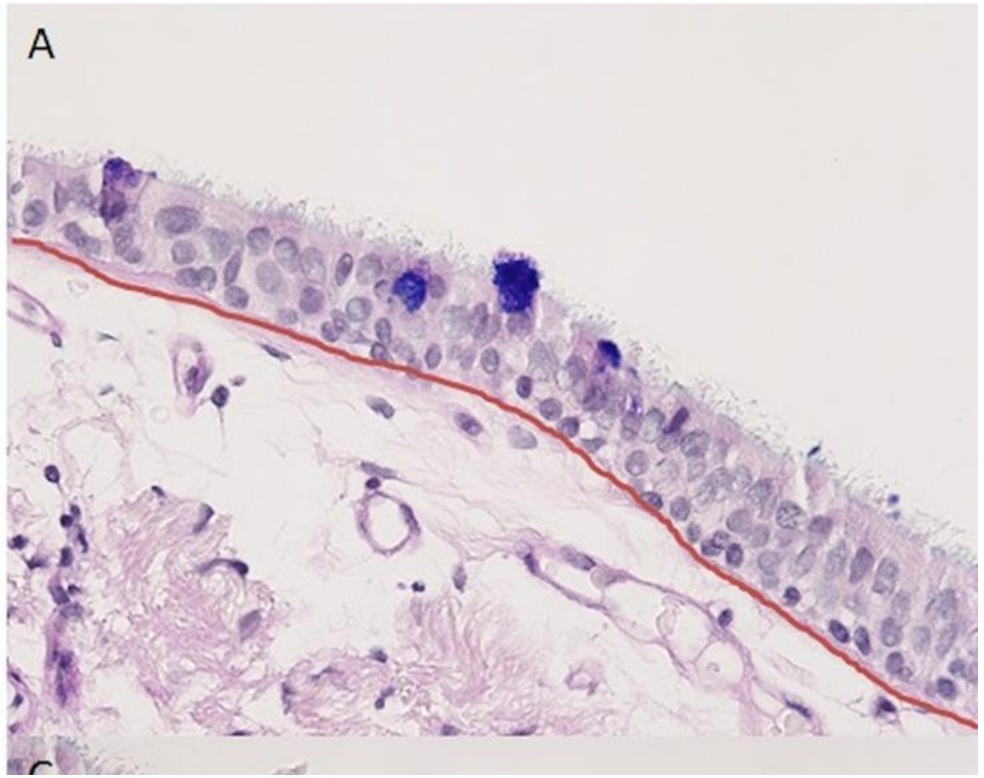


COPD pathophysiology

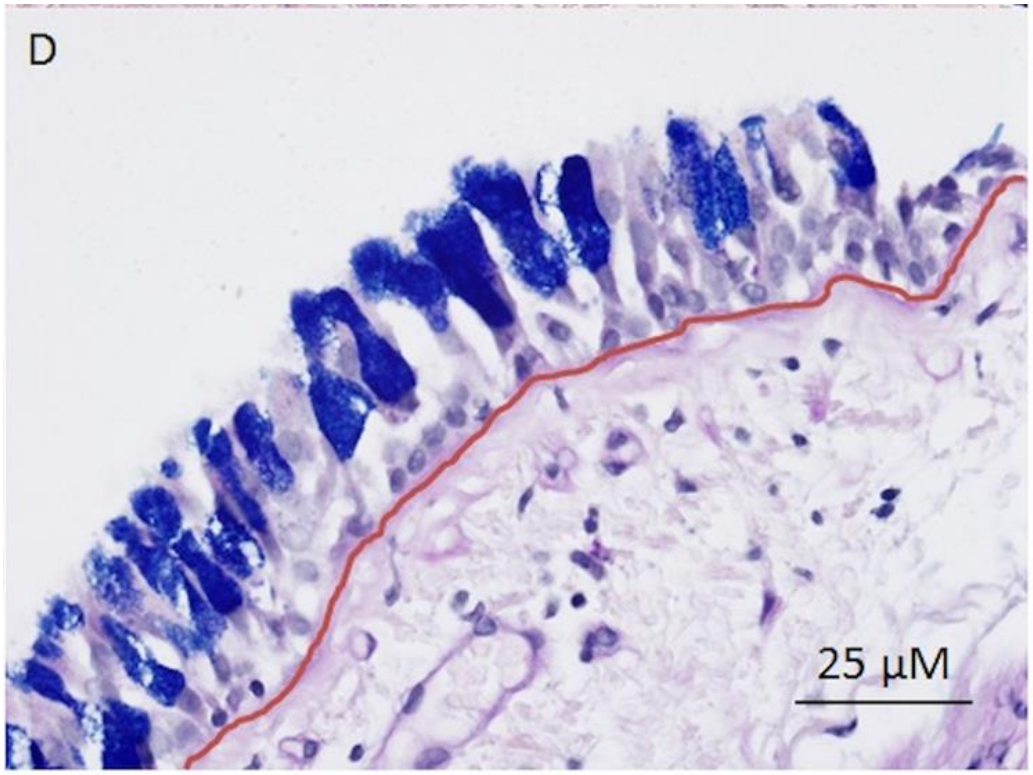


Goblet cell hyperplasia and chronic bronchitis

Healthy airway

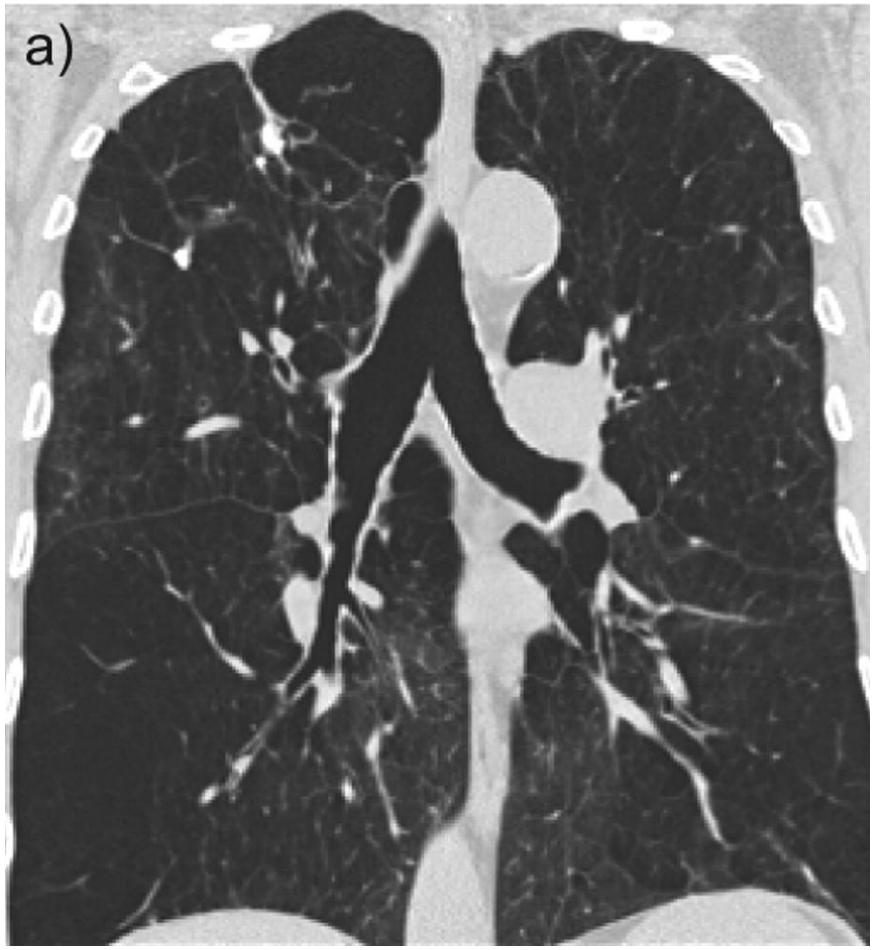


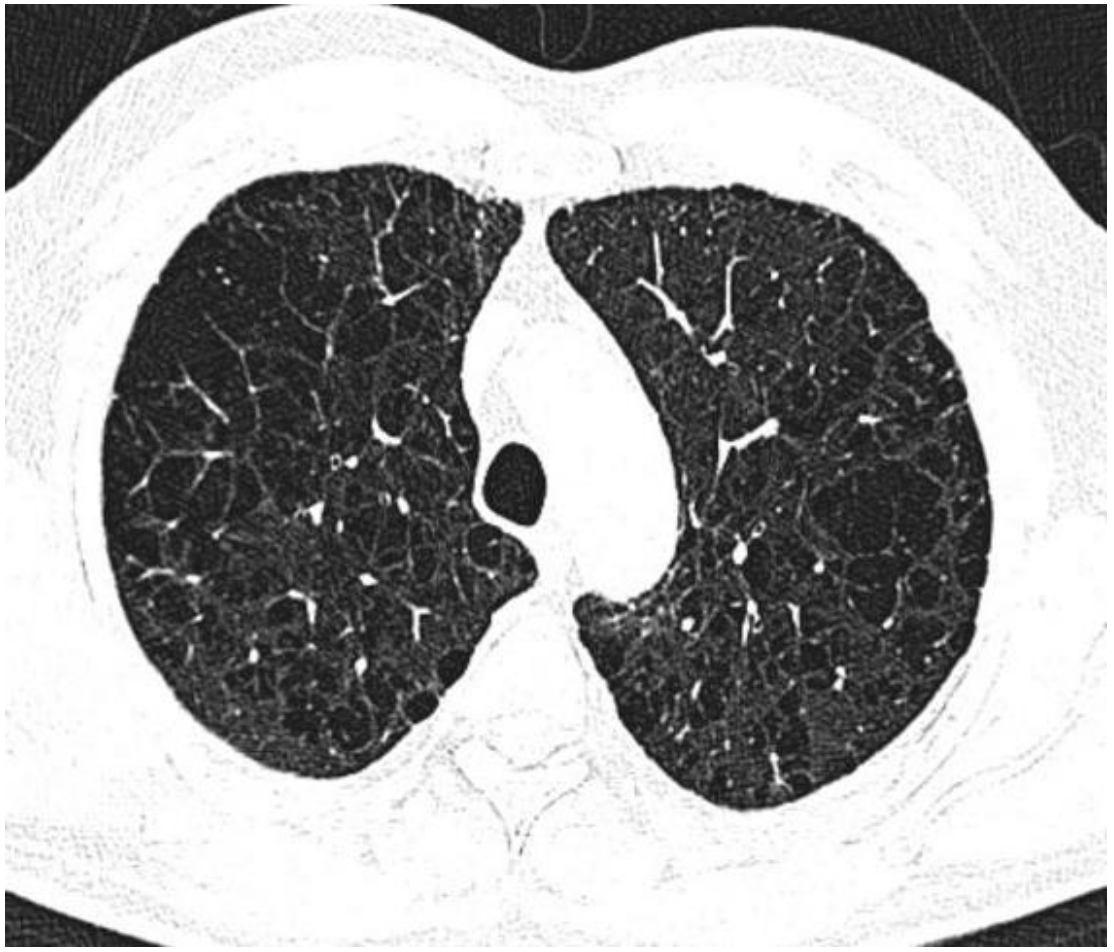
COPD airway



Kim V *et al* Plos One 2015

Emphysema – alveolar destruction





Refaeley Y *et al* ERJ 2010

When should I think about COPD?

- Breathlessness which is
  - Progressive
  - Worse on exercise
- Recurrent wheeze
- Chronic cough
- Recurrent chest infections
- Risk factors



COPD

Making a diagnosis

Air flow obstruction which is not fully reversible

Ndd one Spirometer

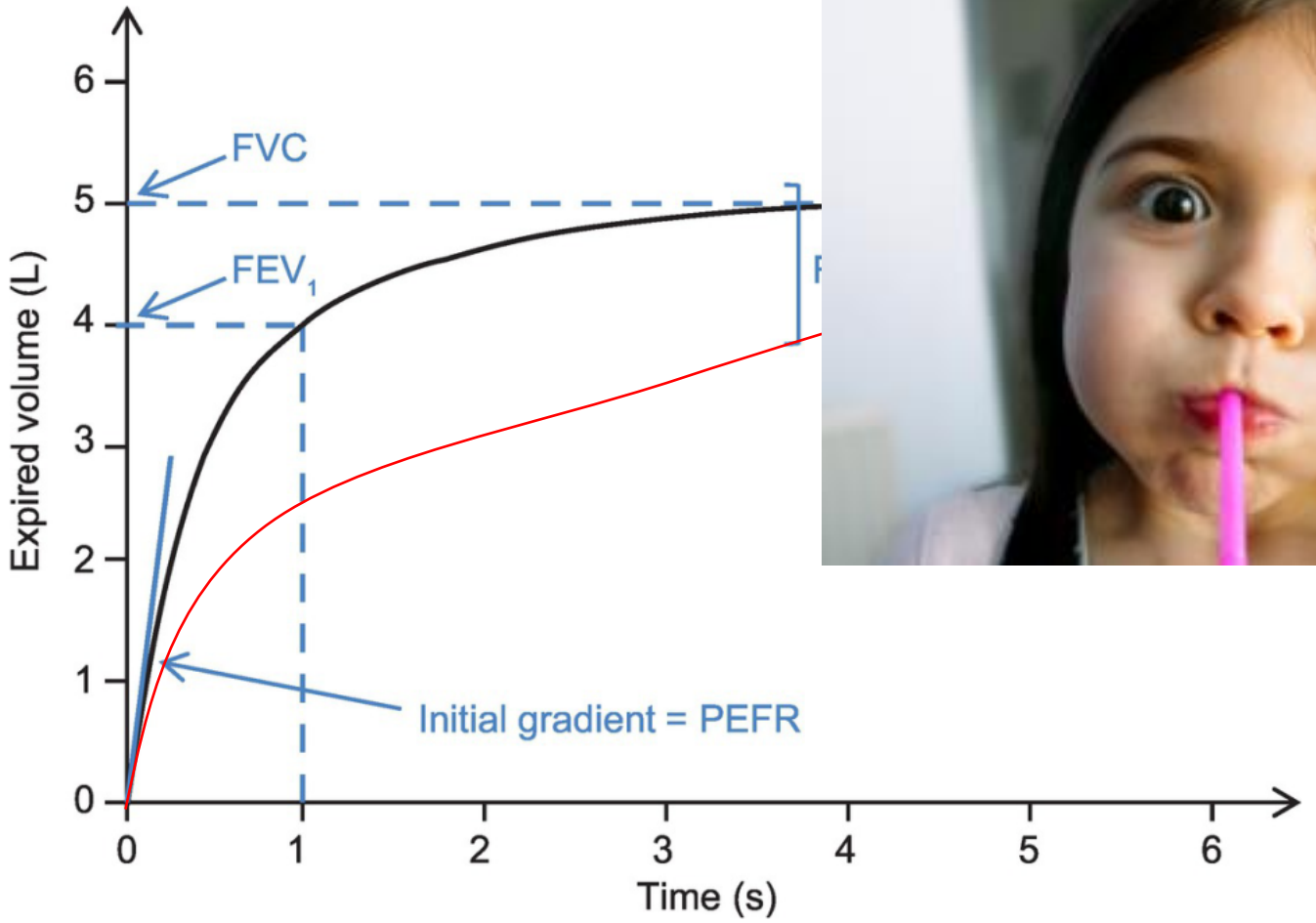


Micro 1 Spirometer

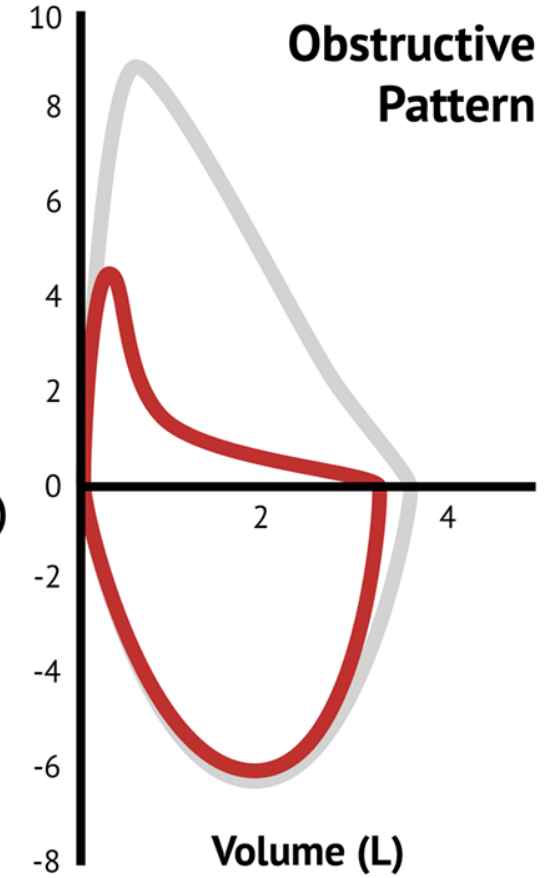


Spirometry

FEV1/FVC<0.70







Grading everity of air flow obstruction

GOLD Grades and Severity of Airflow Obstruction in COPD (based on post-bronchodilator FEV1)

Figure 2.8

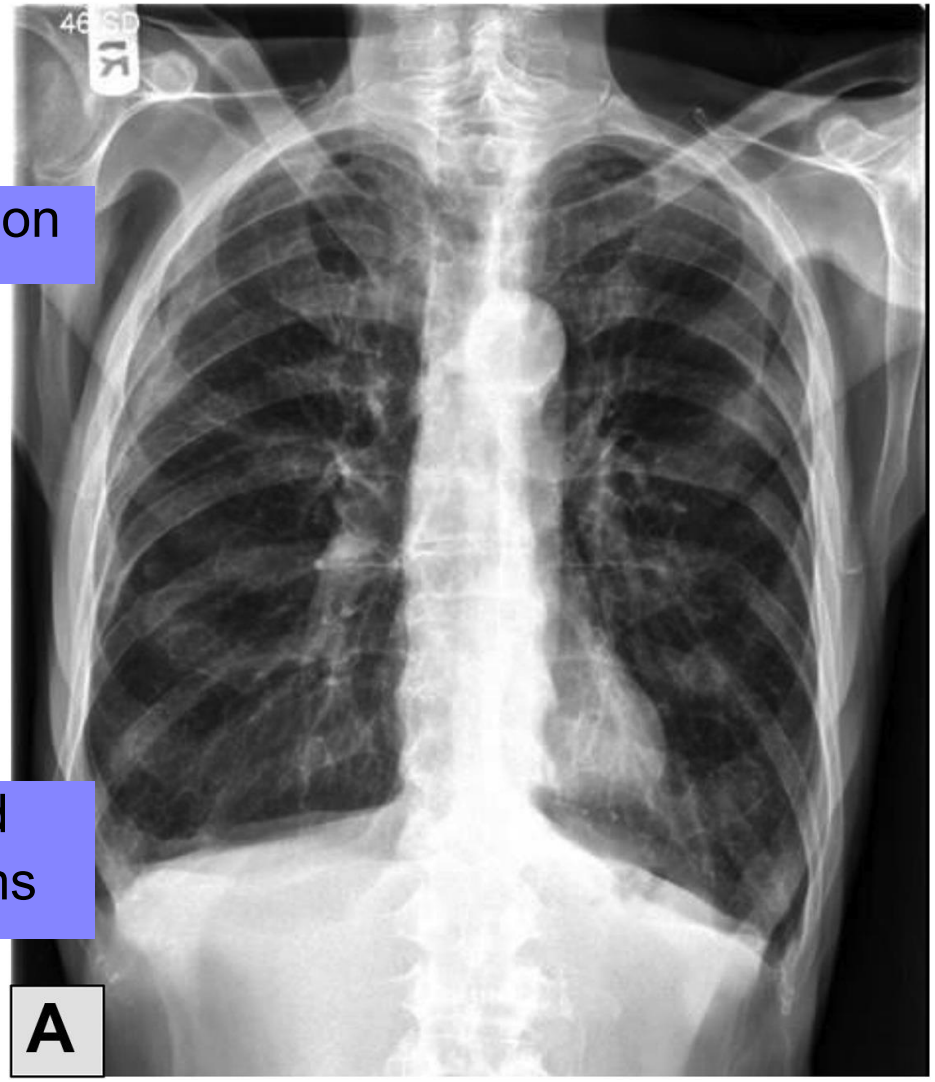
In COPD patients (FEV1/FVC < 0.7):

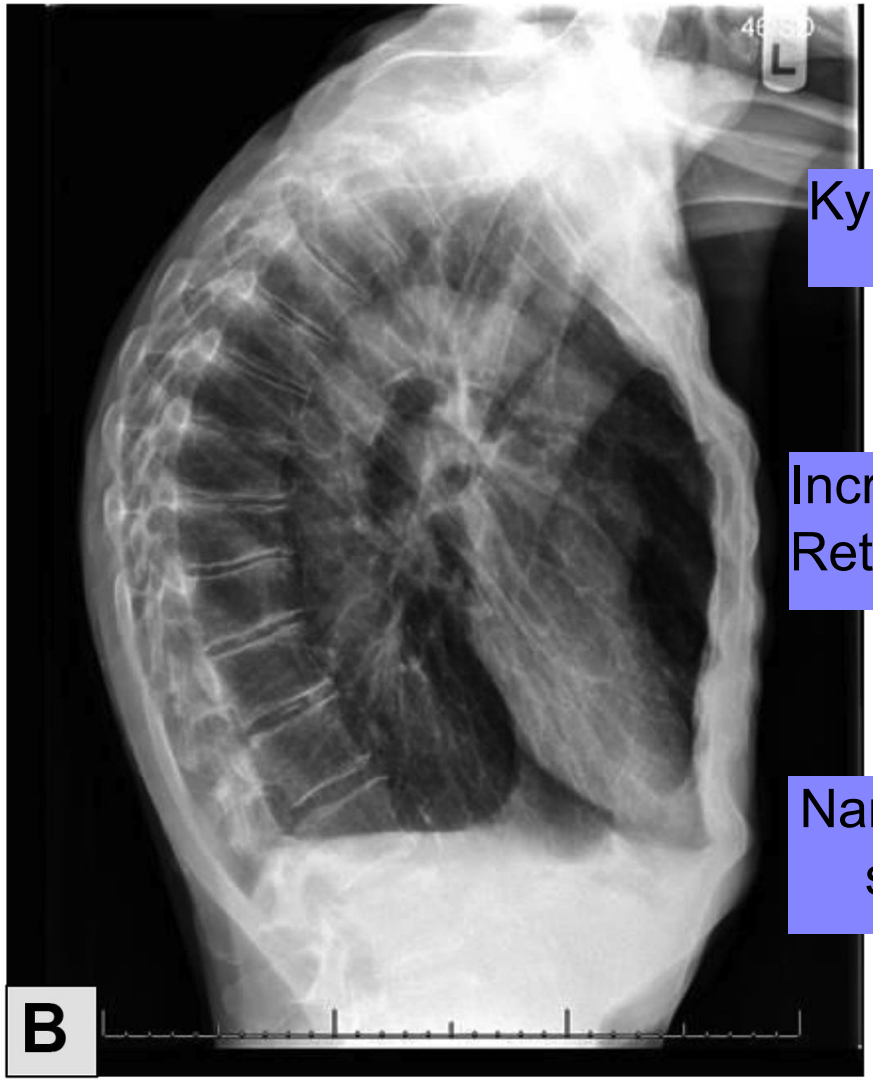
GOLD 1:	Mild	FEV1 ≥ 80% predicted
GOLD 2:	Moderate	50% ≤ FEV1 < 80% predicted
GOLD 3:	Severe	30% ≤ FEV1 < 50% predicted
GOLD 4:	Very Severe	FEV1 < 30% predicted

Chest X-ray in COPD

Hyperinflation

Flattened diaphragms





Kyphoscoliosis

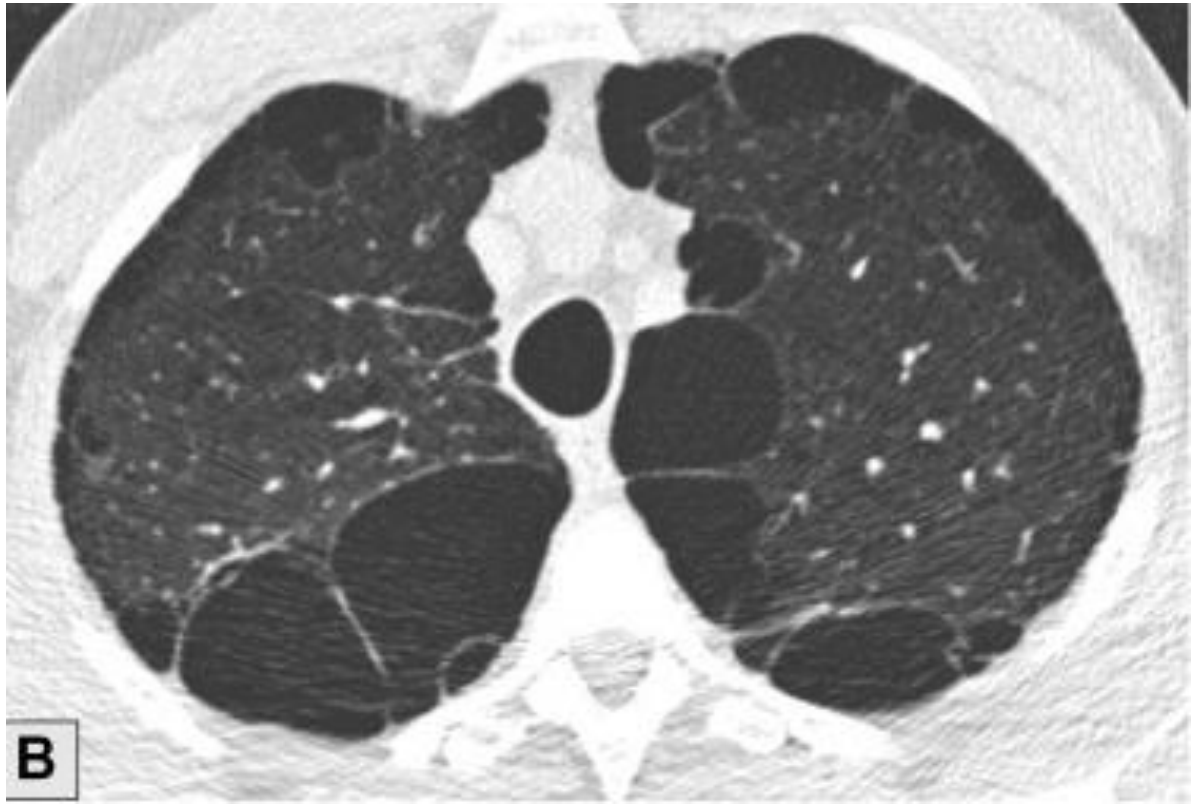
Increased Retro-sternal space

Narrow cardiac silhouette

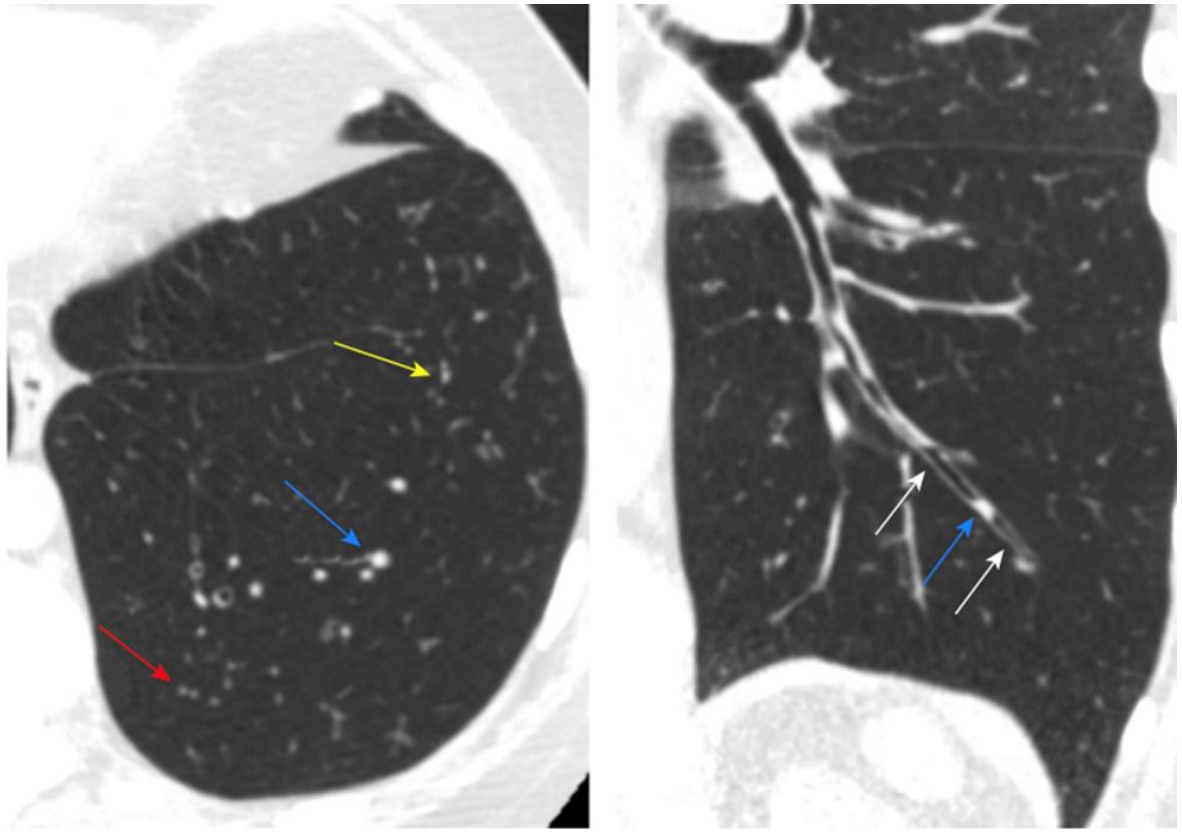
Raoof Chest 2023

CT imaging in COPD

Emphysema



Mucus plugging



Raoof Chest 2023, Diaz JAMA 2024

How bad is it Doc?

Assessing disease severity

Modified MRC dyspnoea scale

Scale	Severity of dyspnoea
0	No breathlessness except with strenuous exercise
1	Shortness of breath when hurrying on the level or walking up a slight hill
2	Walks slower than people of the same age on the level because of breathlessness or has to stop for breath when walking at own pace on the level
3	Stops for breath after walking about 100 metres or after a few minutes on the level
4	Too breathless to leave the house or breathless when dressing or undressing

COPD Assessment Test (CAT)

CAT™ ASSESSMENT

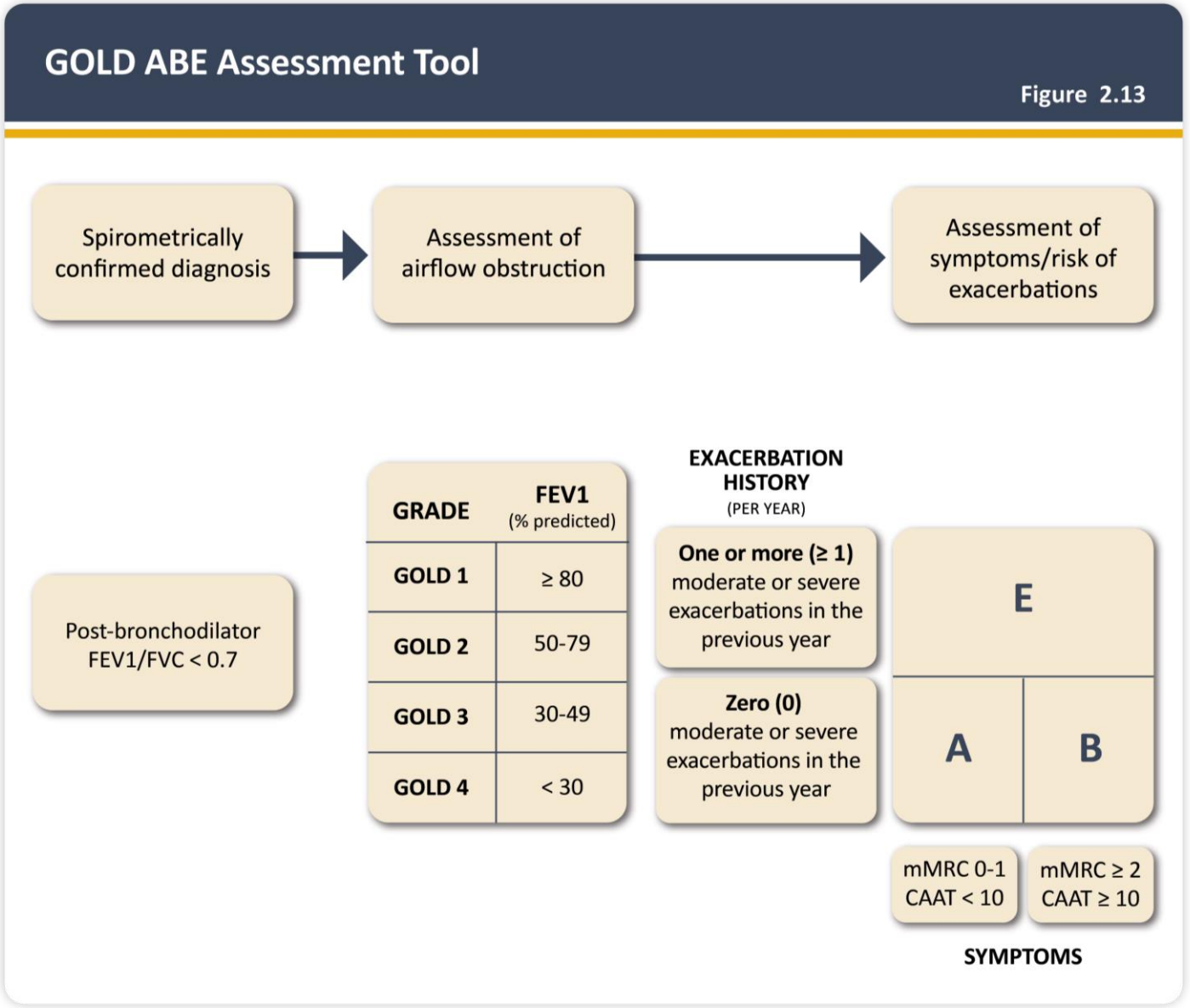
For each item below, place a mark (x) in the box that best describes you currently.
Be sure to only select one response for each question.

EXAMPLE: I am very happy	0 (X) 2 3 4 5	**I am very sad**	**SCORE**
I never cough	0 1 2 3 4 5	I cough all the time	
I have no phlegm (mucus) in my chest at all	0 1 2 3 4 5	My chest is completely full of phlegm (mucus)	
My chest does not feel tight at all	0 1 2 3 4 5	My chest feels very tight	
When I walk up a hill or one flight of stairs I am not breathless	0 1 2 3 4 5	When I walk up a hill or one flight of stairs I am very breathless	
I am not limited doing any activities at home	0 1 2 3 4 5	I am very limited doing activities at home	
I am confident leaving my home despite my lung condition	0 1 2 3 4 5	I am not at all confident leaving my home because of my lung condition	
I sleep soundly	0 1 2 3 4 5	I don't sleep soundly because of my lung condition	
I have lots of energy	0 1 2 3 4 5	I have no energy at all	

Reference: Jones et al. ERJ 2009; 34 (3); 648-54.

TOTAL SCORE: ( )

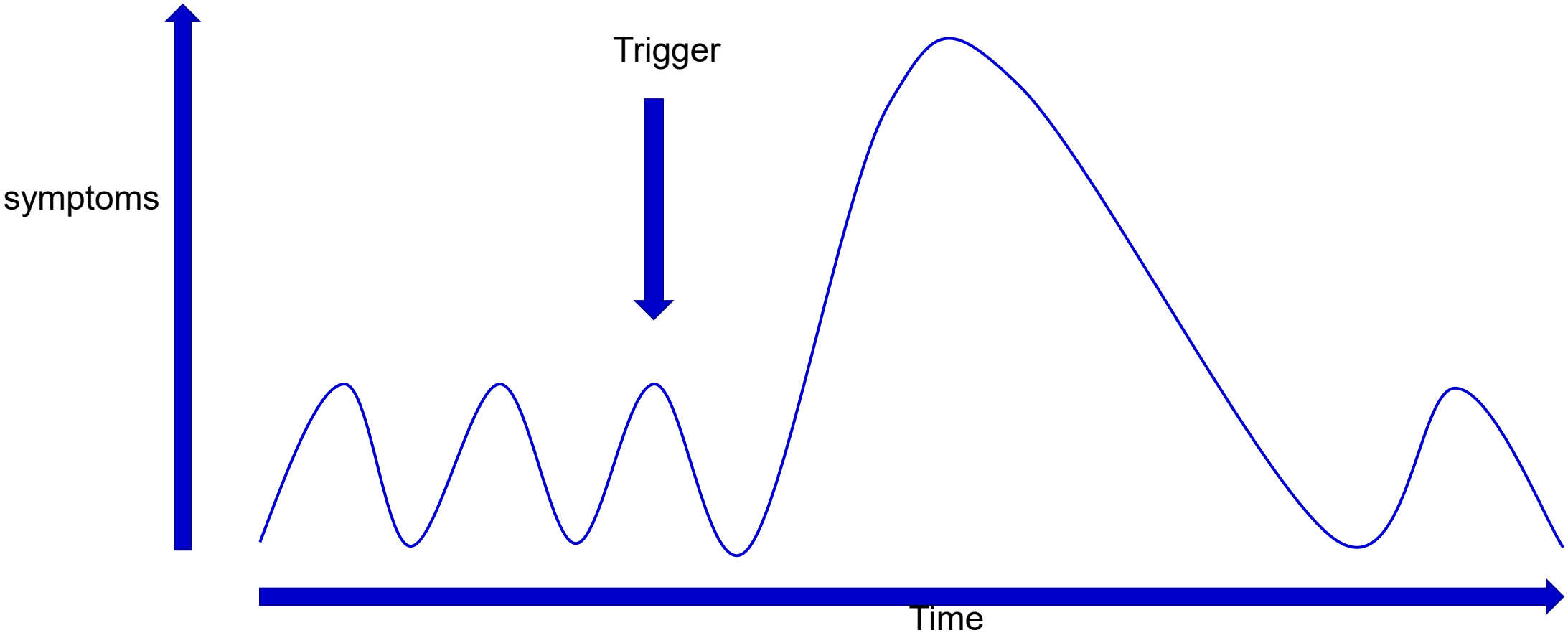
GOLD Assessment of COPD



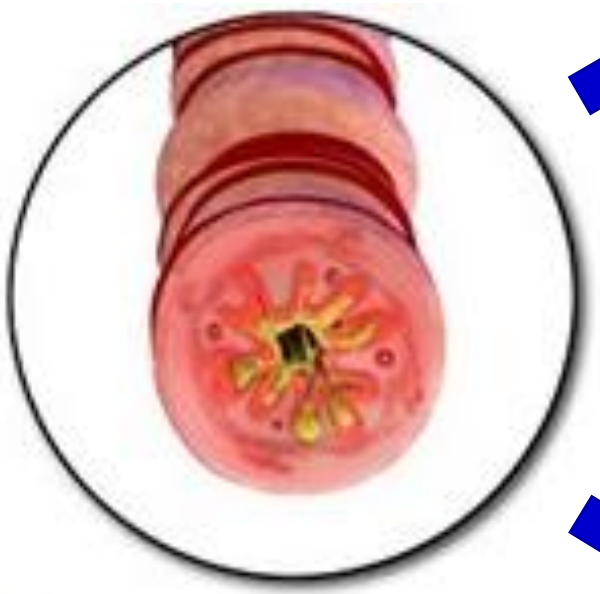
COPD Exacerbations

Exacerbation definition

“An acute event characterised by increased cough/sputum/ breathlessness worsening over 14 days” GOLD 2025



Symptoms of COPD exacerbations



Airway inflammation

Mucus hypersecretion

Bronchial constriction

Dynamic hyperinflation

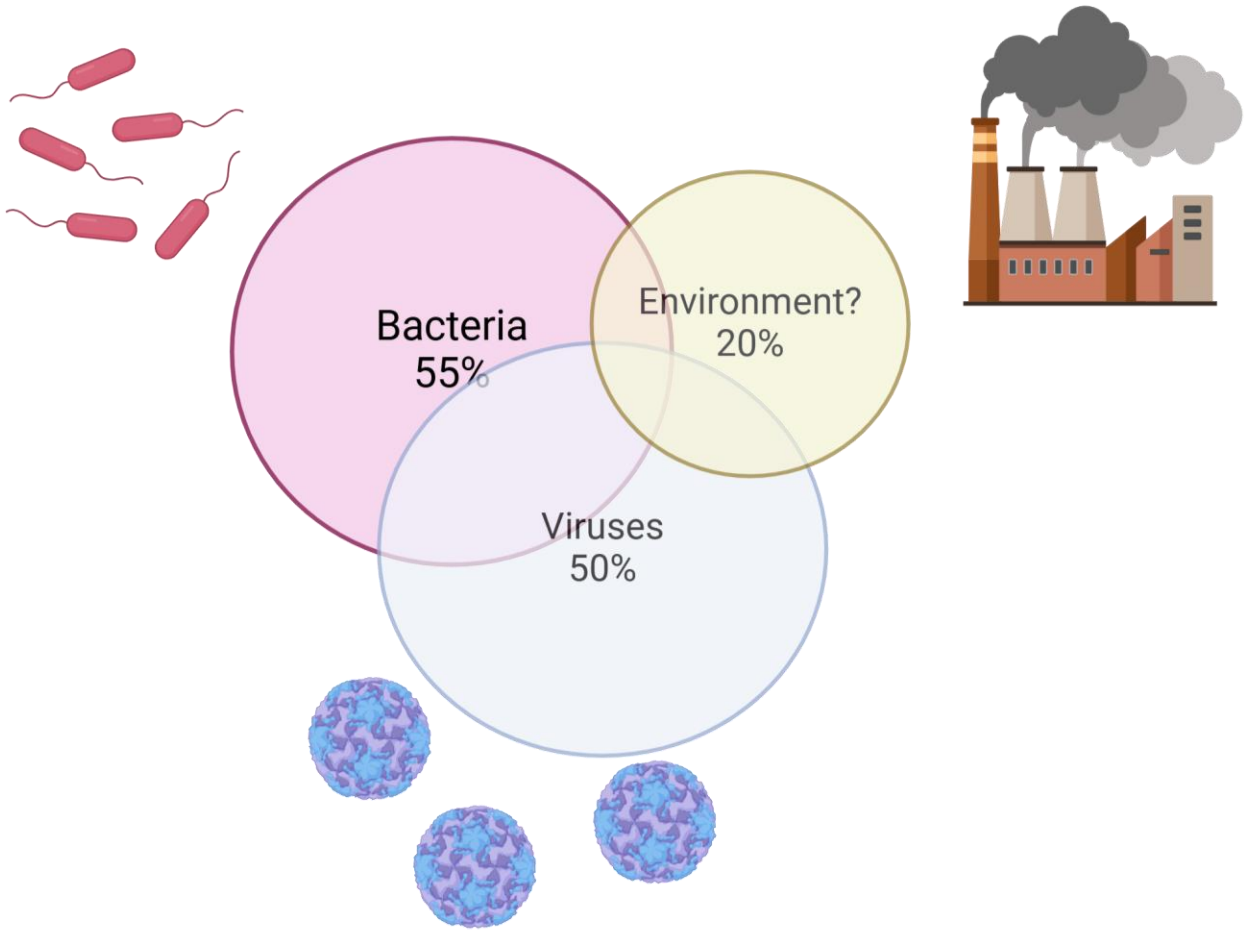
Anthonisen criteria

↑ Sputum volume

↑ Sputum purulence

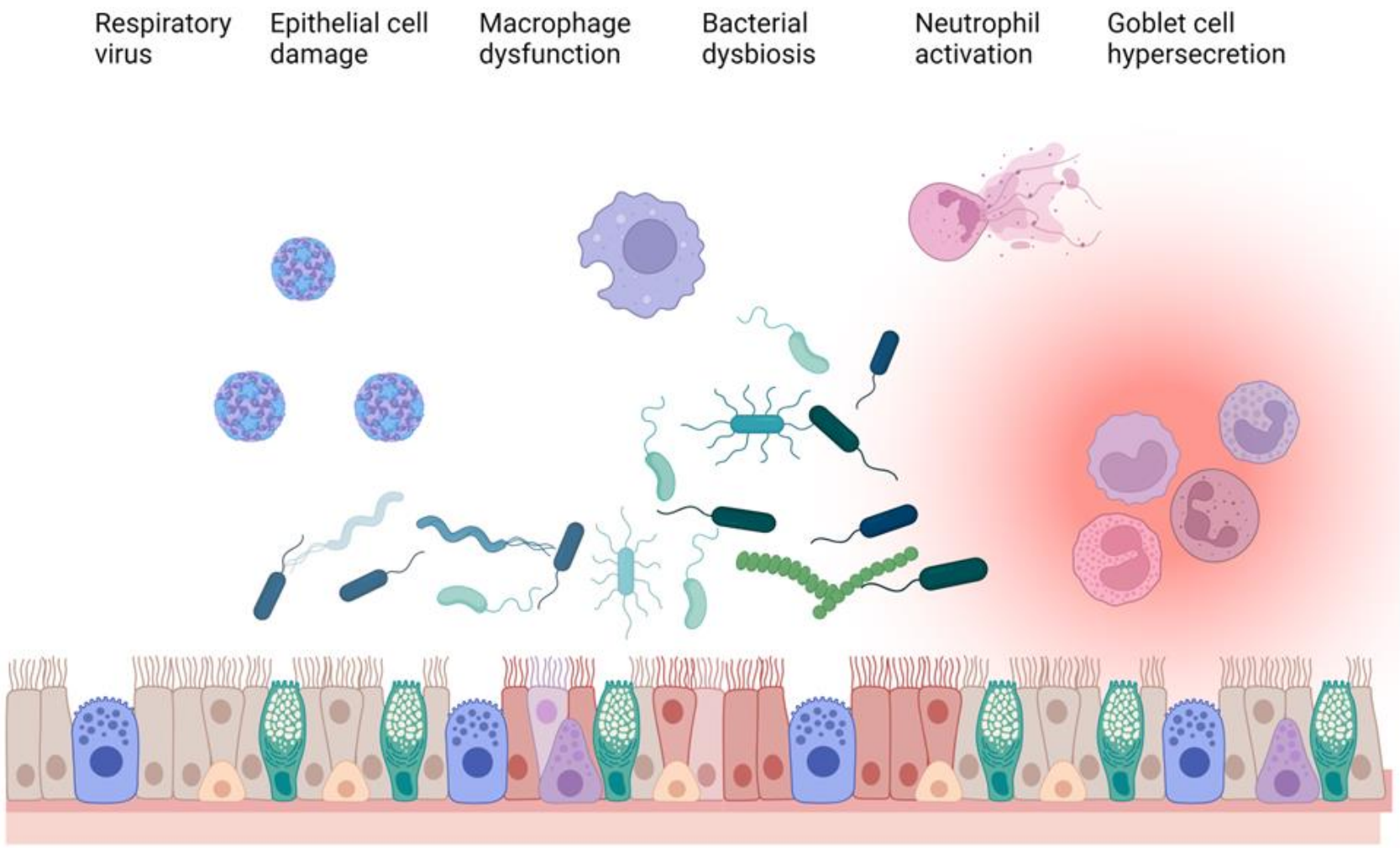
↑Breathlessness

Causes of severe exacerbations



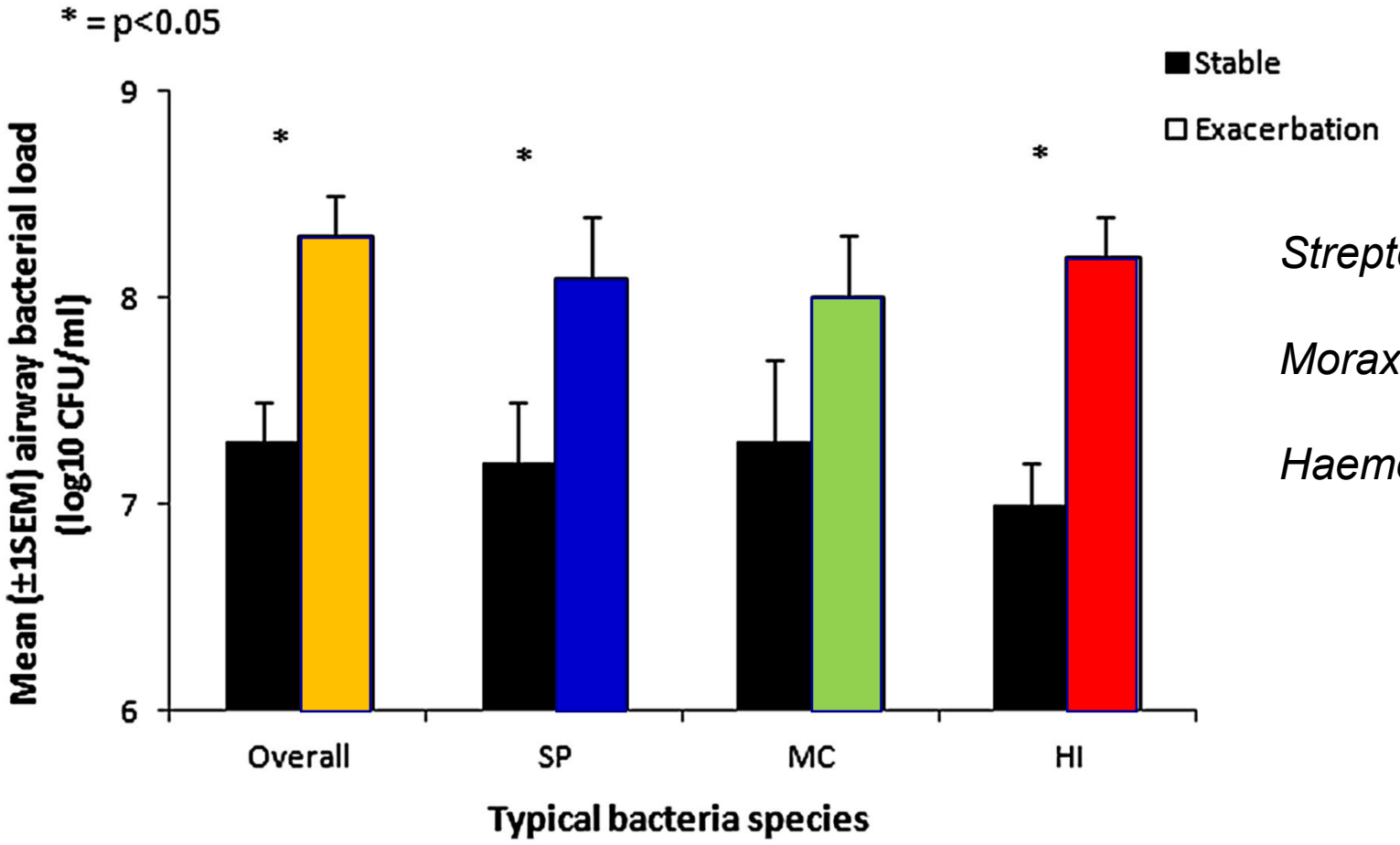
Papi AJRCCM 2006:17;1114

COPD Exacerbation Pathogenesis



Finney ERJ COPD Monograph 2024

Bacteria in COPD exacerbations



Streptococcus pneumoniae

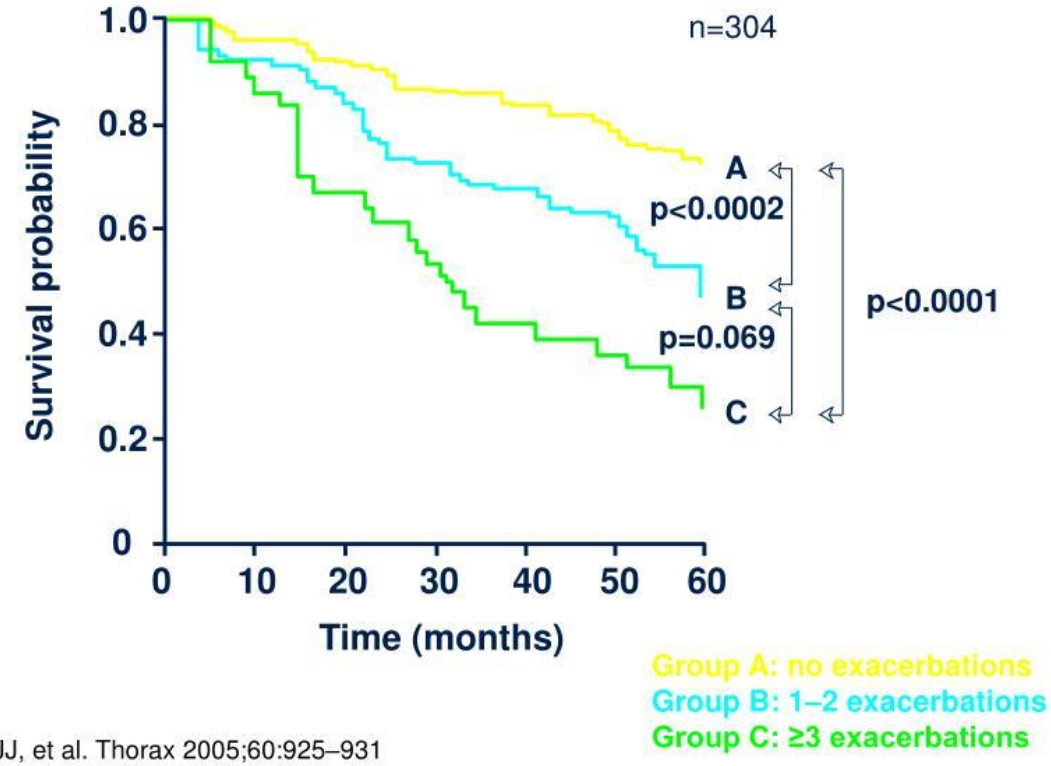
Moraxella catarrhalis

Haemophilus influenzae

Garcha *et al* Thorax 2012

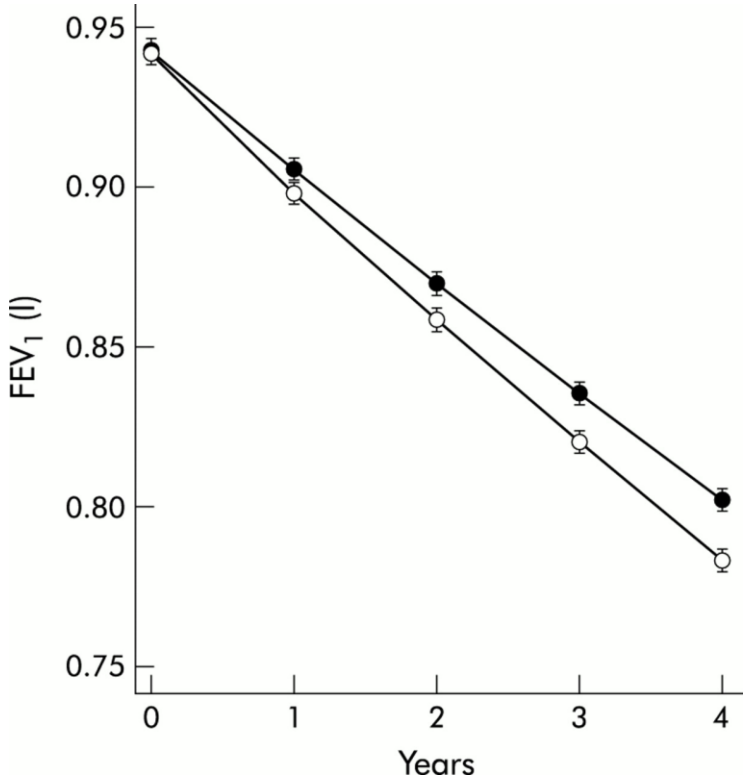
The problem with COPD exacerbations

The impact of frequent COPD exacerbations - more frequent attacks increase mortality



Soler-Cataluna JJ, et al. Thorax 2005;60:925–931

COPD exacerbations are associated with rapid lung function decline

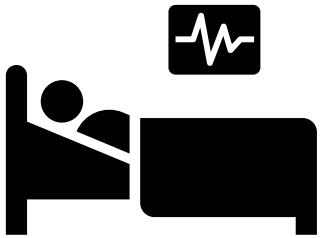


Donaldson et al. Thorax 2002

Mortality from severe COPD exacerbations

NIV 8% mortality

In hospital 5%



90 day 10%



Readmission
↑ 30% mortality

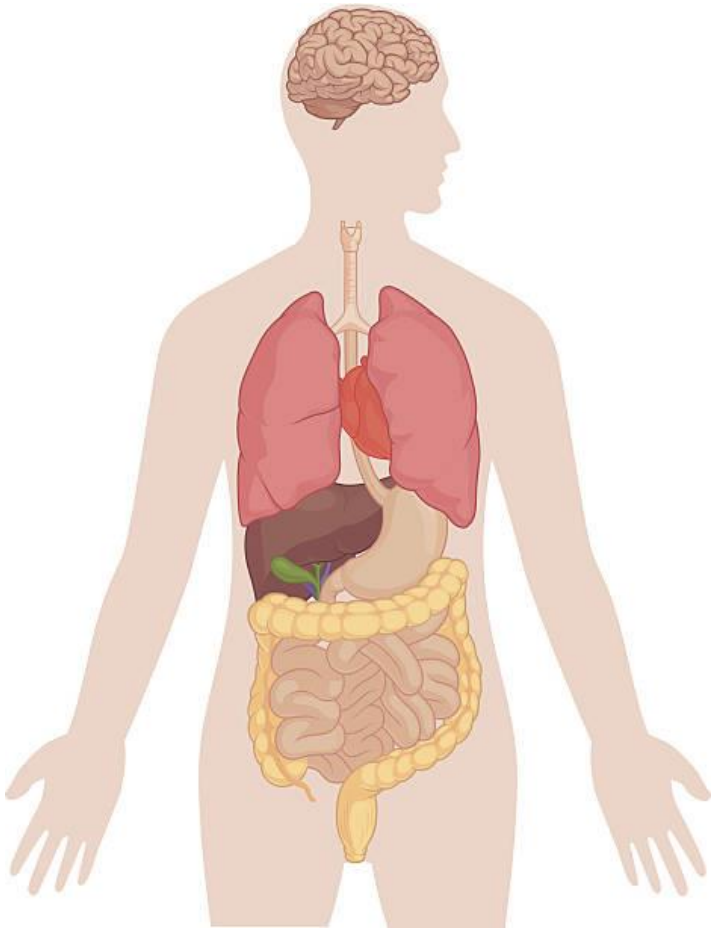
COPD UK National Audit

Co-morbidities
Systemic factors associated with increased risk of exacerbation

Anxiety or depression

Gastroesophageal reflux

Female sex



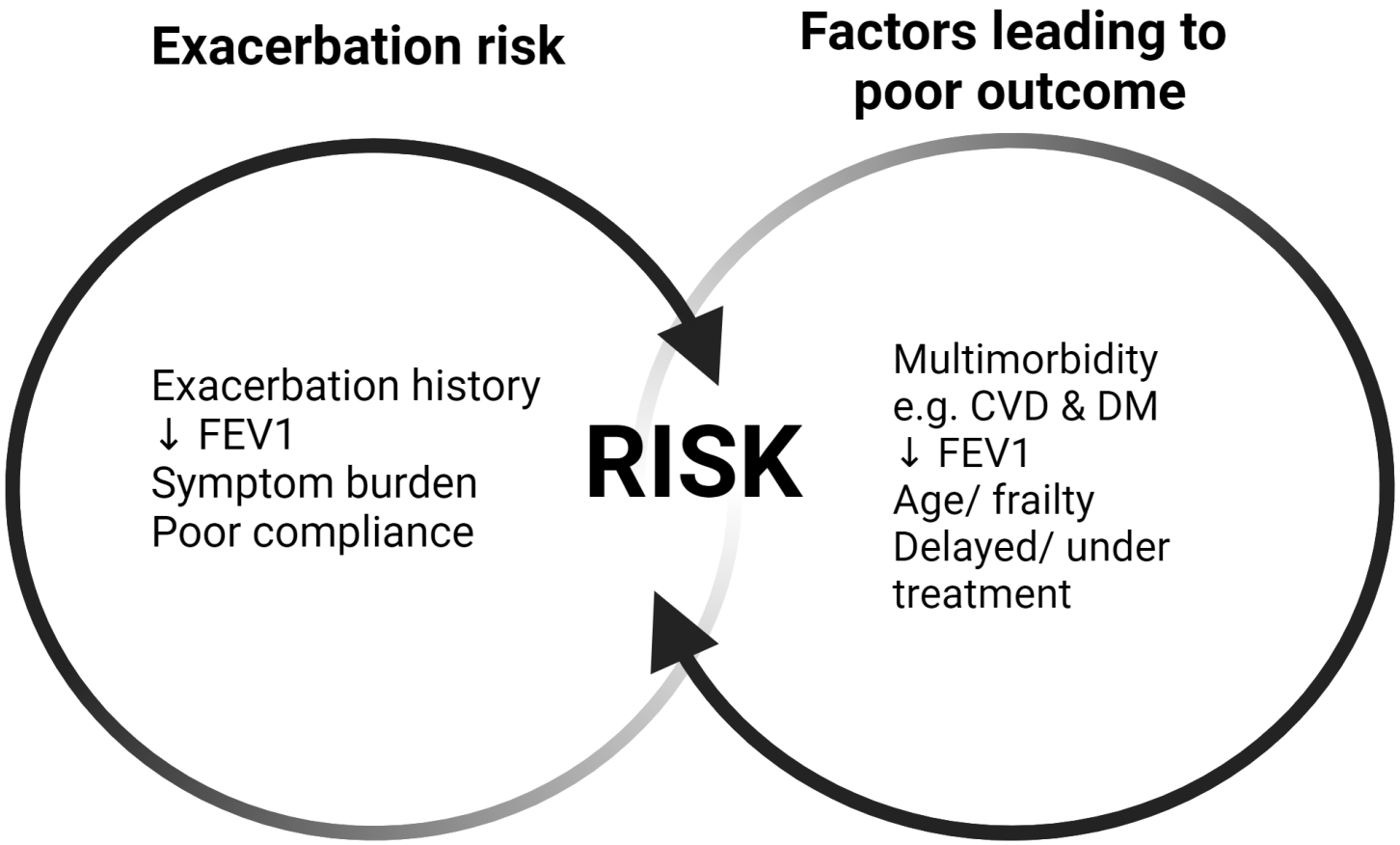
Systemic inflammation

Cardiovascular disease

Diabetes

Augusti *et al* COPD 2014

Exacerbation Risk



Finney *et al* COPD ERS Monograph 2024

Initial assessment

Severity	Thresholds
Mild	RR≤24 HR≤95 bpm Sats >92% or at baseline Breathlessness VAS<5
Moderate (3 or more)	RR>24 HR>95 bpm Sats <92% or ≥3% change from baseline CRP>10 Not acidotic on ABG Breathlessness VAS ≥ 5
Severe	ABG shows acidosis pH<7.35 and CO2 >6 kPa/ 45 mmHg

The Rome Proposal. Celli *et al* AJRCCM 2021: 204 (11);1245

Acute respiratory failure
GOLD guidelines 2026

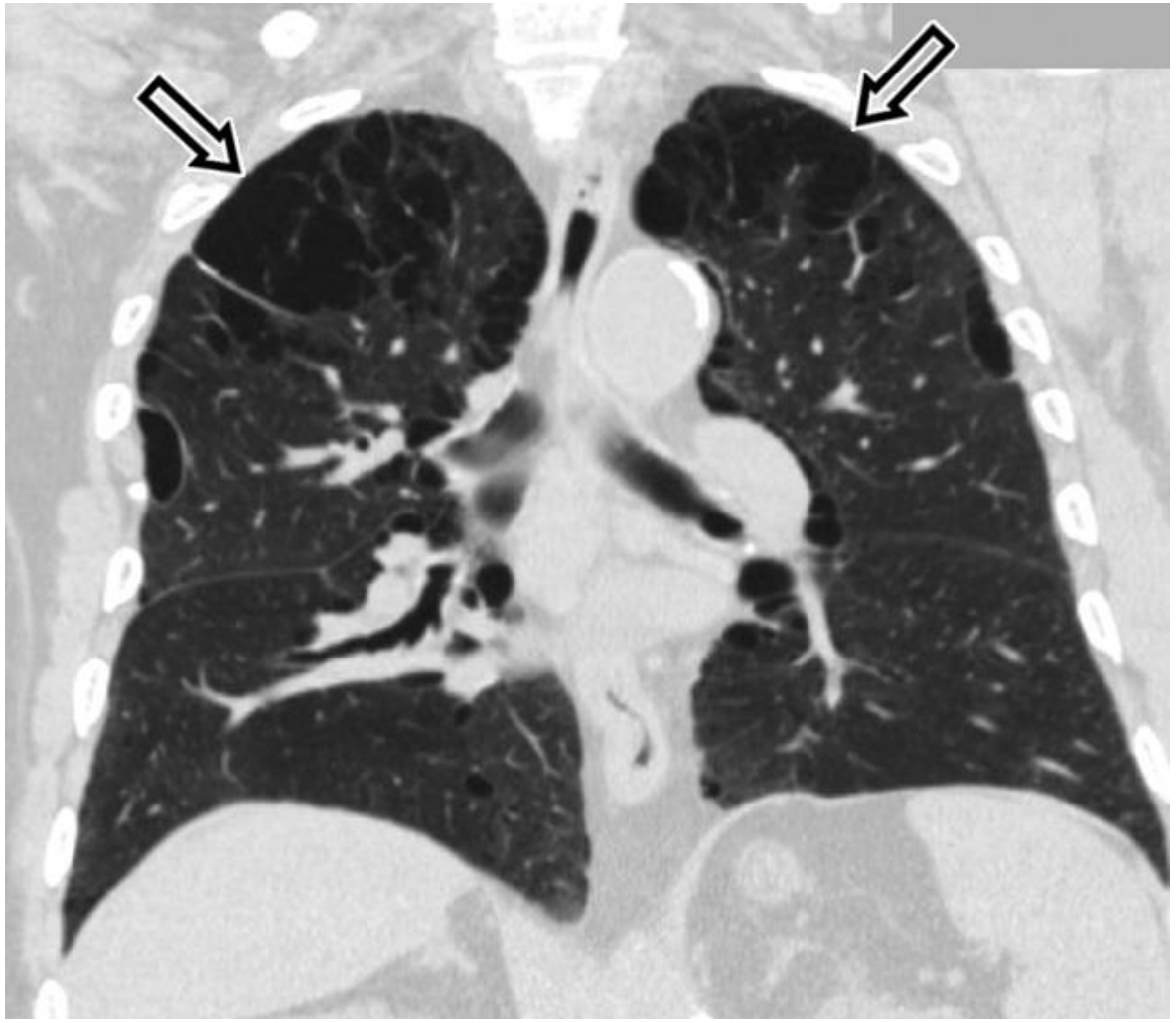
Acute respiratory failure – non-life threatening

- RR>24, accessory muscle use,
- FiO2≤35%, ↑CO2 but not acidotic

Acute respiratory failure – life threatening

- RR>24, accessory muscle use,
- Change in mental status
- Hypoxia FiO2 ≥ 40%
- ↑CO2 ≥8 kPa or acidotic pH<7.35

Controlled oxygen therapy



- Consider baseline sats if known
- Target sats 88-92%
- V/Q mismatch
- Remember! Repeat ABG

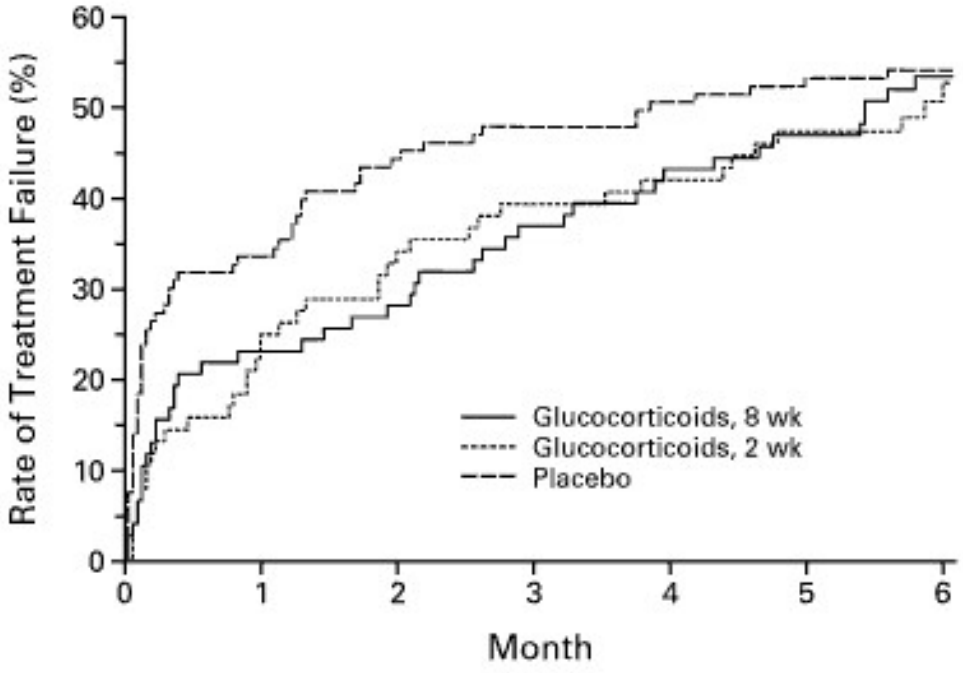
Bronchodilators



- Salbutamol MDI via spacer 10 puffs
- Or Salbutamol 2.5 mg first line

- Greatest bronchodilator effect
- Can keep usual maintenance inhalers

Oral corticosteroids



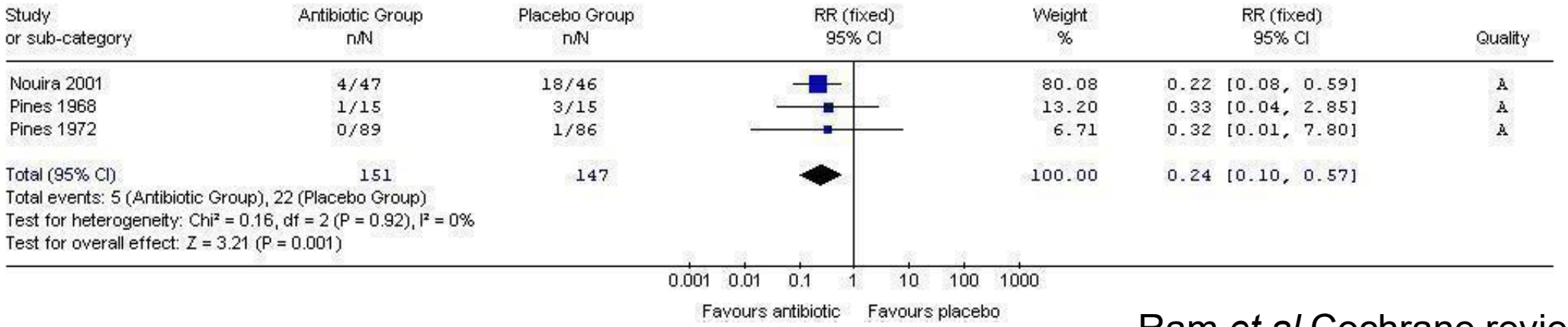
No. at Risk				
Glucocorticoids, 8 wk	80	61	50	21
Glucocorticoids, 2 wk	80	59	46	20
Placebo	111	74	58	39

- Reduce length of stay
- Improve FEV1
- Reduce risk of treatment failure (Readmission)
- Prolonged treatment associated with pneumonia/ hyperglycaemia
- Increase bacterial load
- 30 mg Prednisolone for 5 days

Nielwenhoer *et al* NEJM 1999: 340;1941

Antibiotics

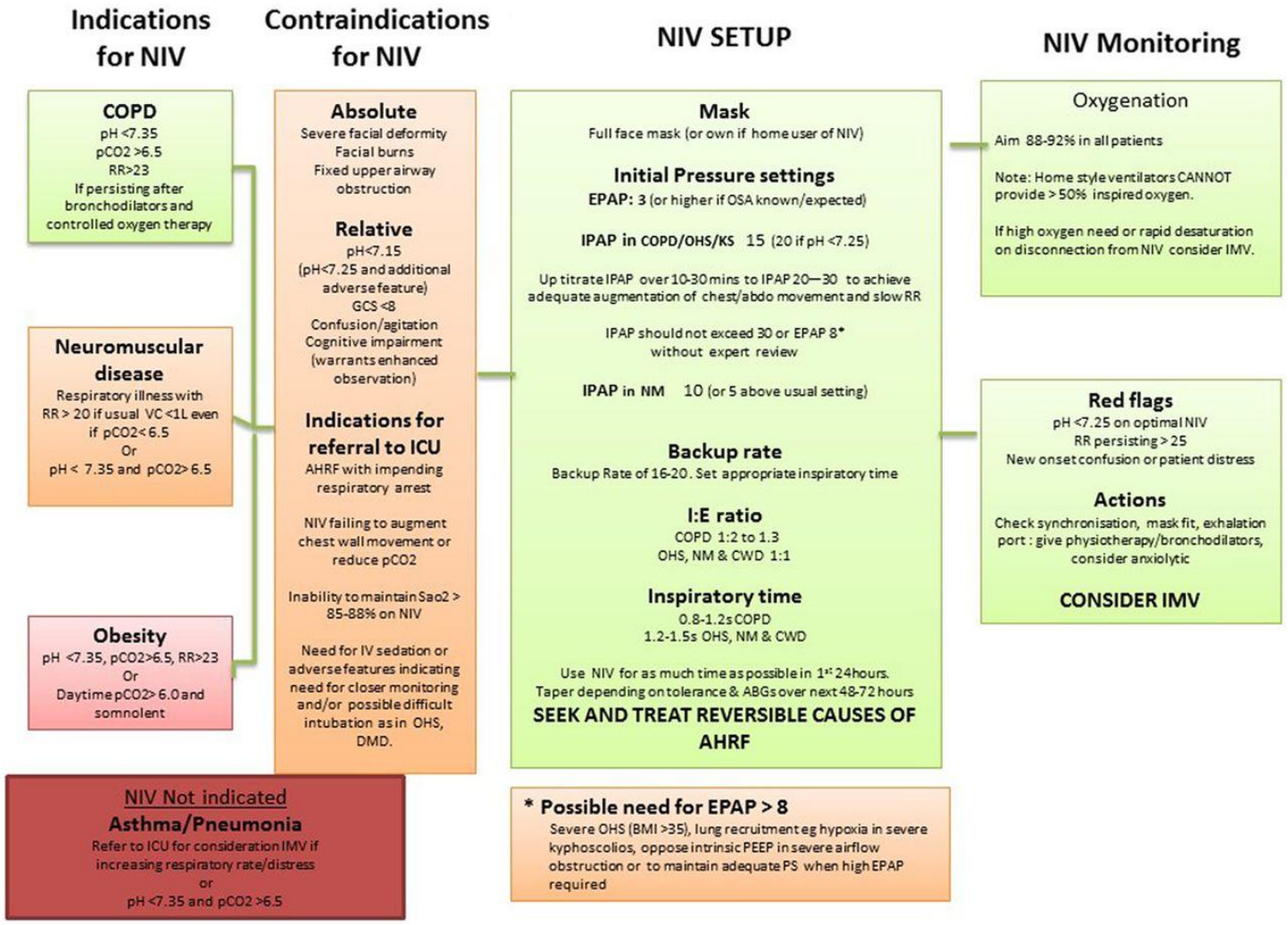
Review: *Antibiotics for exacerbations of chronic obstructive pulmonary disease
Comparison: 01 Antibiotics versus Placebo
Outcome: 01 Mortality



Ram *et al* Cochrane review 2006

↓Mortality 77%
↓ Treatment failure 53%
↓Sputum purulence 44%
Beware Pseudomonas/ gram negative organisms

Summary for providing acute non-invasive ventilation.



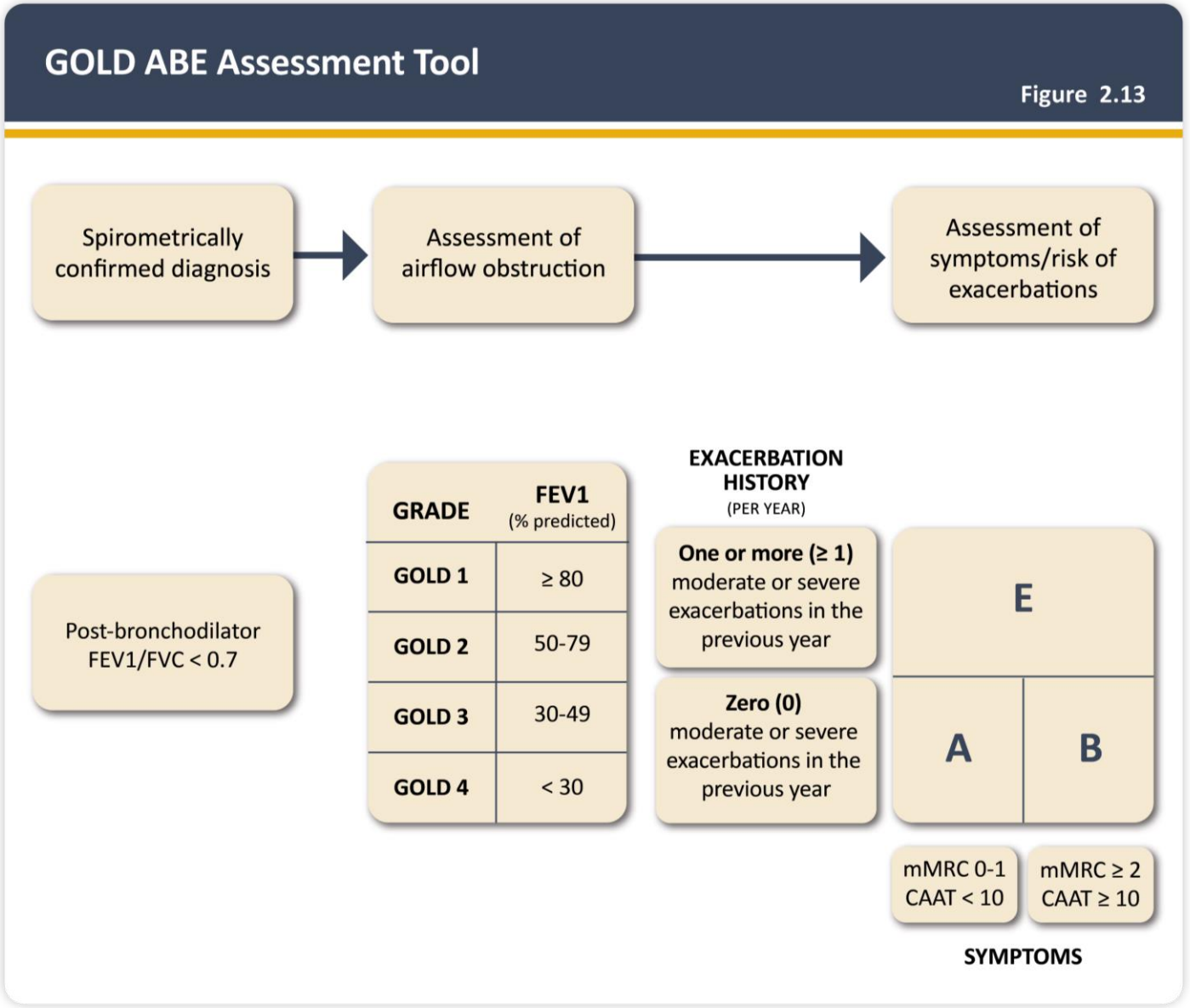
A Craig Davidson et al. Thorax 2016;71:ii1-ii35

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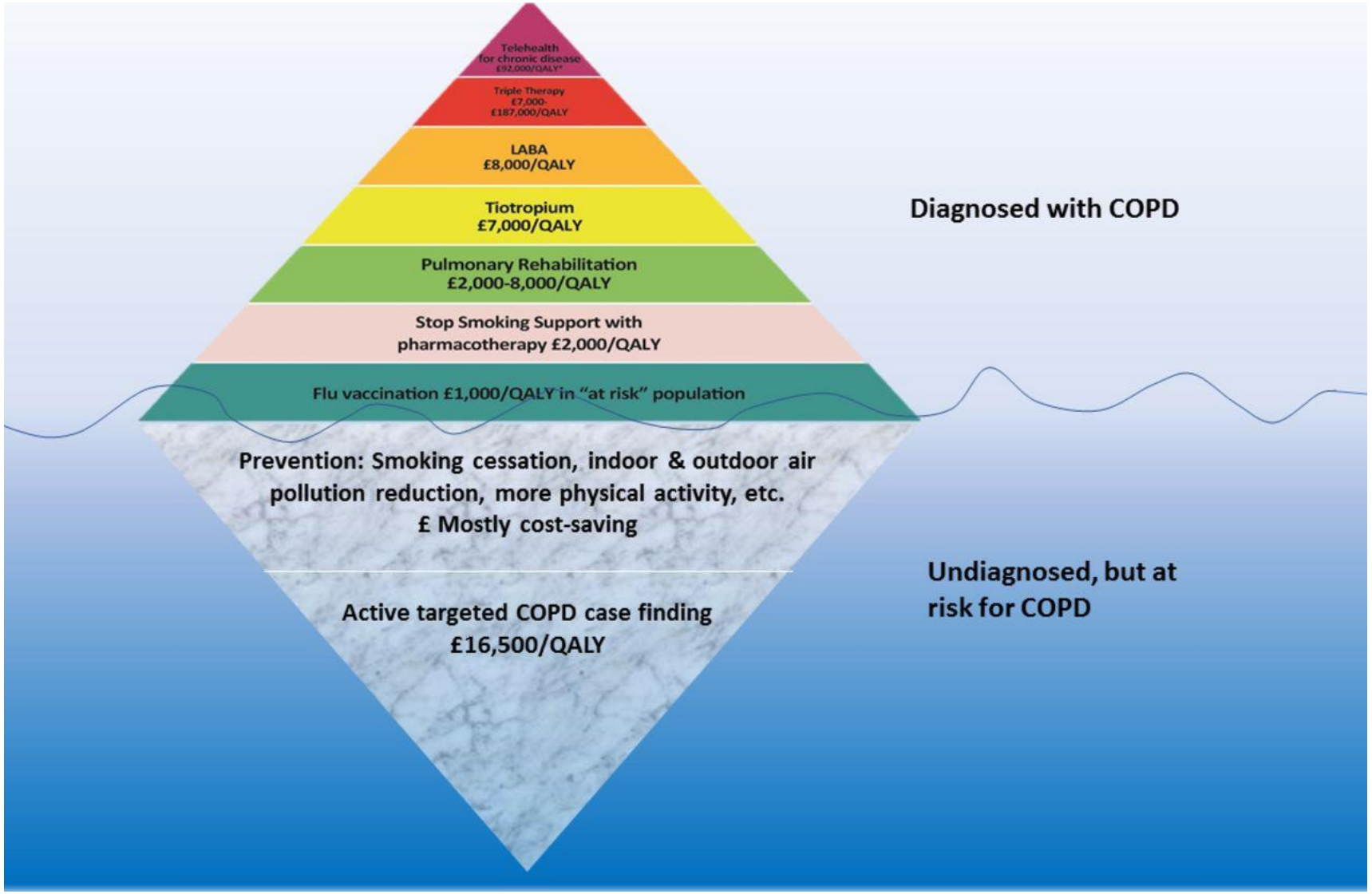
THORAX

Management of stable COPD

GOLD Assessment of COPD

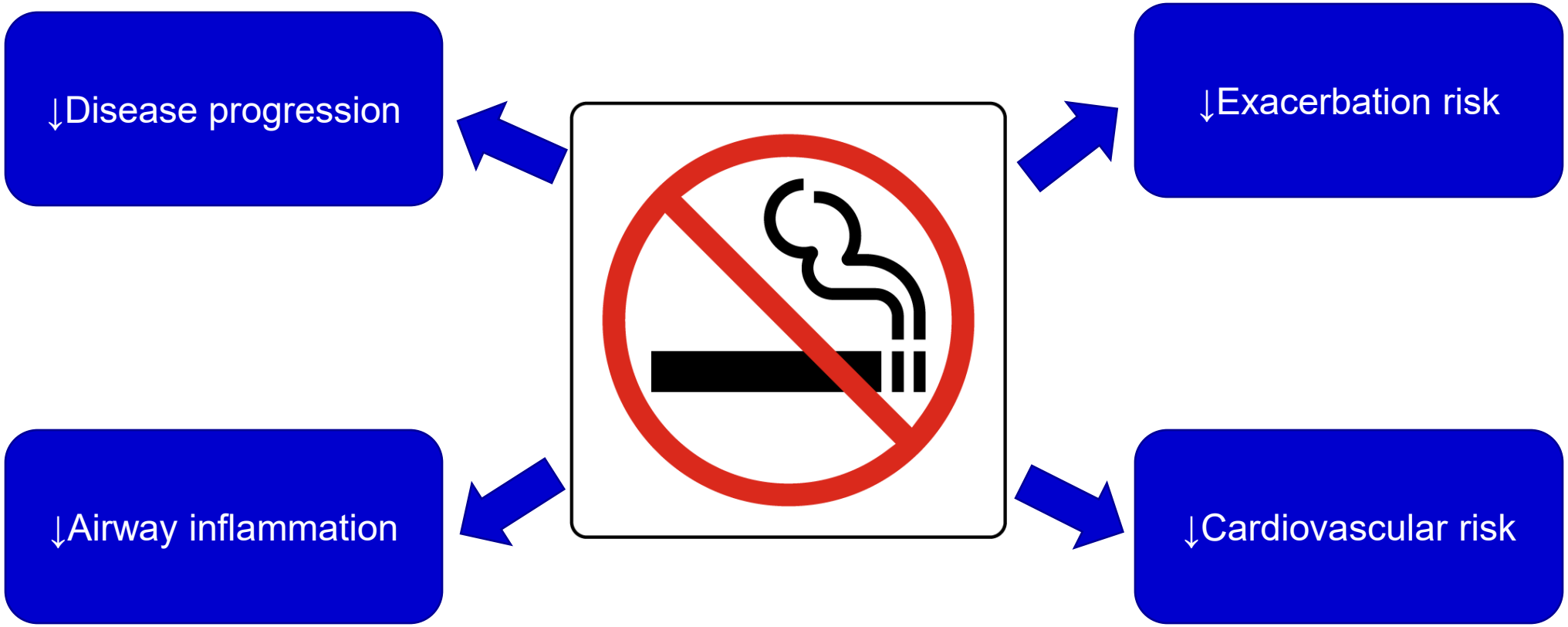


Don’t forget the basics



Boven JFM Thorax 2019; **74:**727-729.

Smoking cessation



Pulmonary Rehabilitation

A PATH TO BREATHING BETTER

People diagnosed with chronic obstructive pulmonary disease (COPD) know that it is a serious and debilitating condition requiring ongoing medical attention. But many do not realize that pulmonary rehabilitation may help them live and breathe better. This therapy is most effective when started in the moderate stage of COPD—but even people who have advanced stage lung disease can benefit.

HELPS TO ▶



Reduce COPD symptoms



Increase physical activity



Improve daily life function



Improve emotional health

Pulmonary rehabilitation

In person assessment

Endurance and resistance exercise



Structured education programme

Assessment at the beginning and end

Man Thorax 2023

What about inhalers?

Bronchodilators

- Short acting
- Beta-agonists
- Relief of breathlessnes

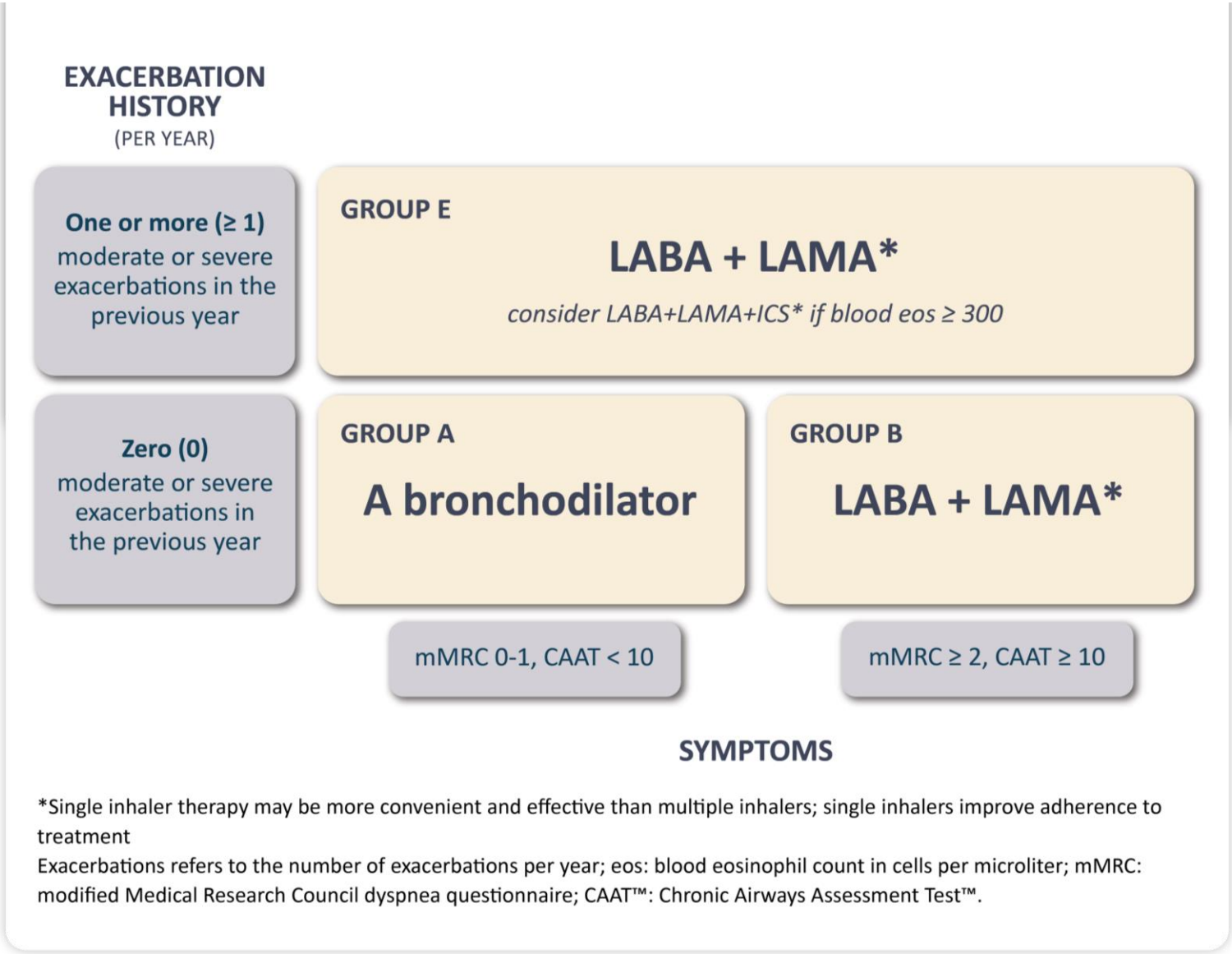
Bronchodilators

- Long-acting
- Beta-agonists
- Anti-muscarinics
- Prevent breathlessnes
- Reduce exacerbations

Inhaled steroids

- Long-acting
- Reduce airway inflammation
- Can increase pneumonia risk
- Blood eosinophils can guide

Initial inhaler treatment for COPD

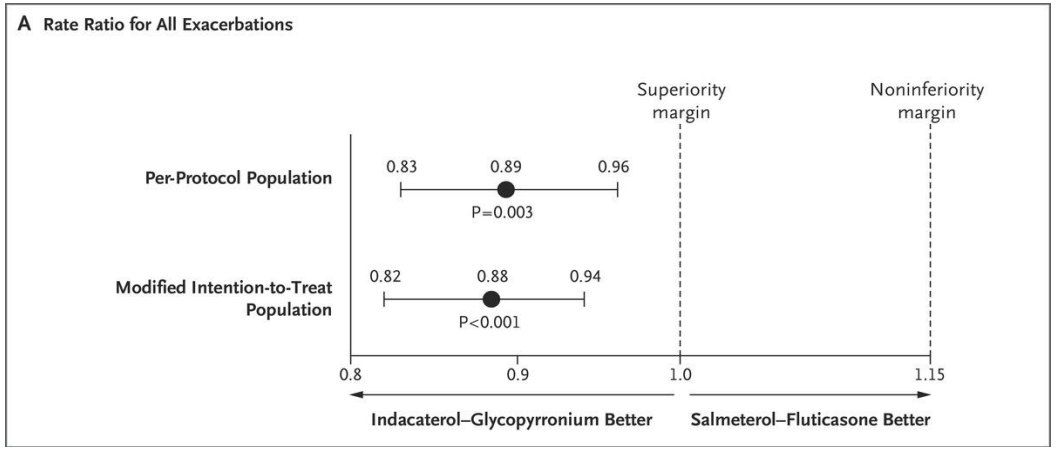


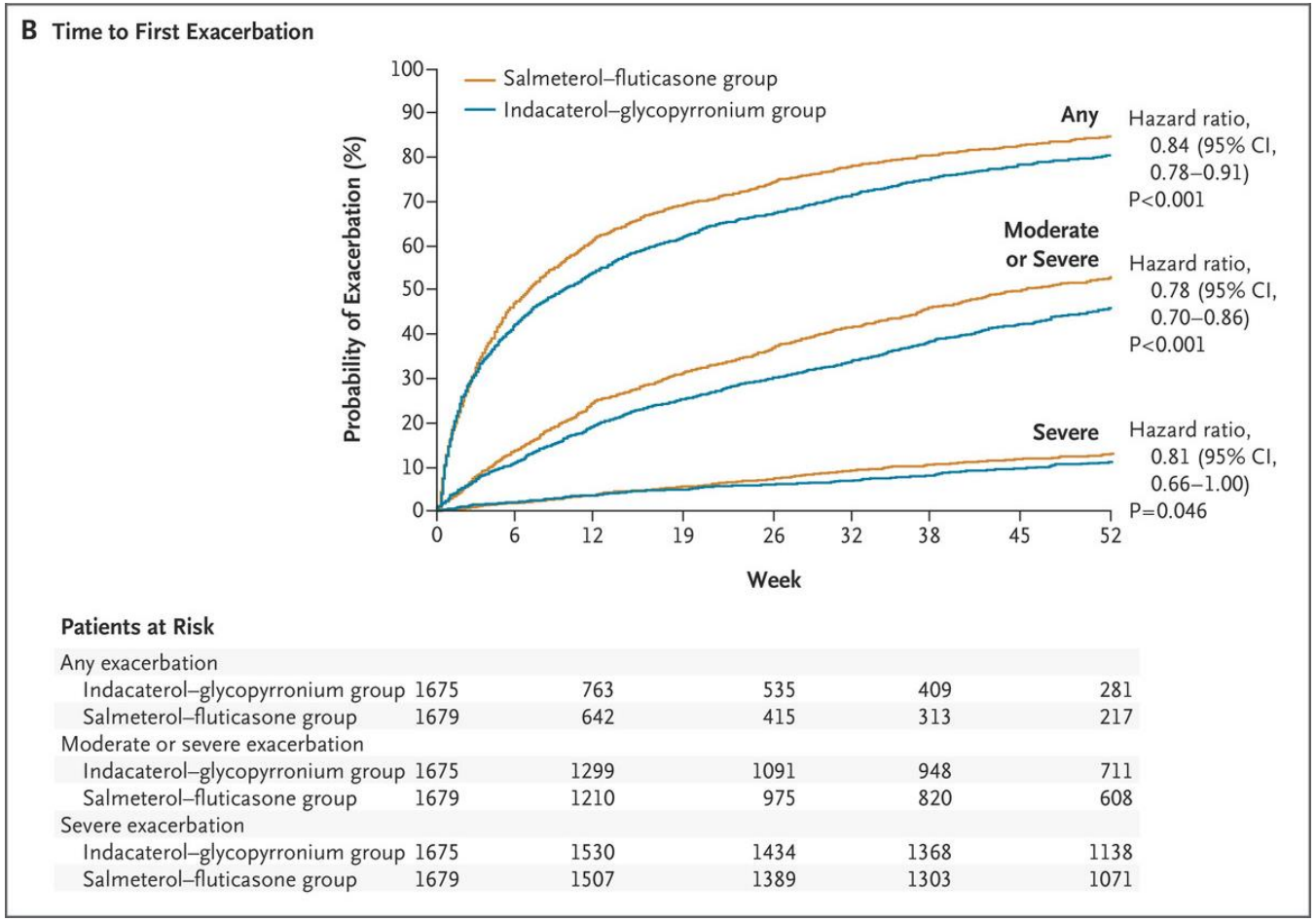
*Single inhaler therapy may be more convenient and effective than multiple inhalers; single inhalers improve adherence to treatment

Exacerbations refers to the number of exacerbations per year; eos: blood eosinophil count in cells per microliter; mMRC: modified Medical Research Council dyspnea questionnaire; CAAT™: Chronic Airways Assessment Test™.

GOLD 2026

FLAME study – LABA/LAMA vs ICS/LABA



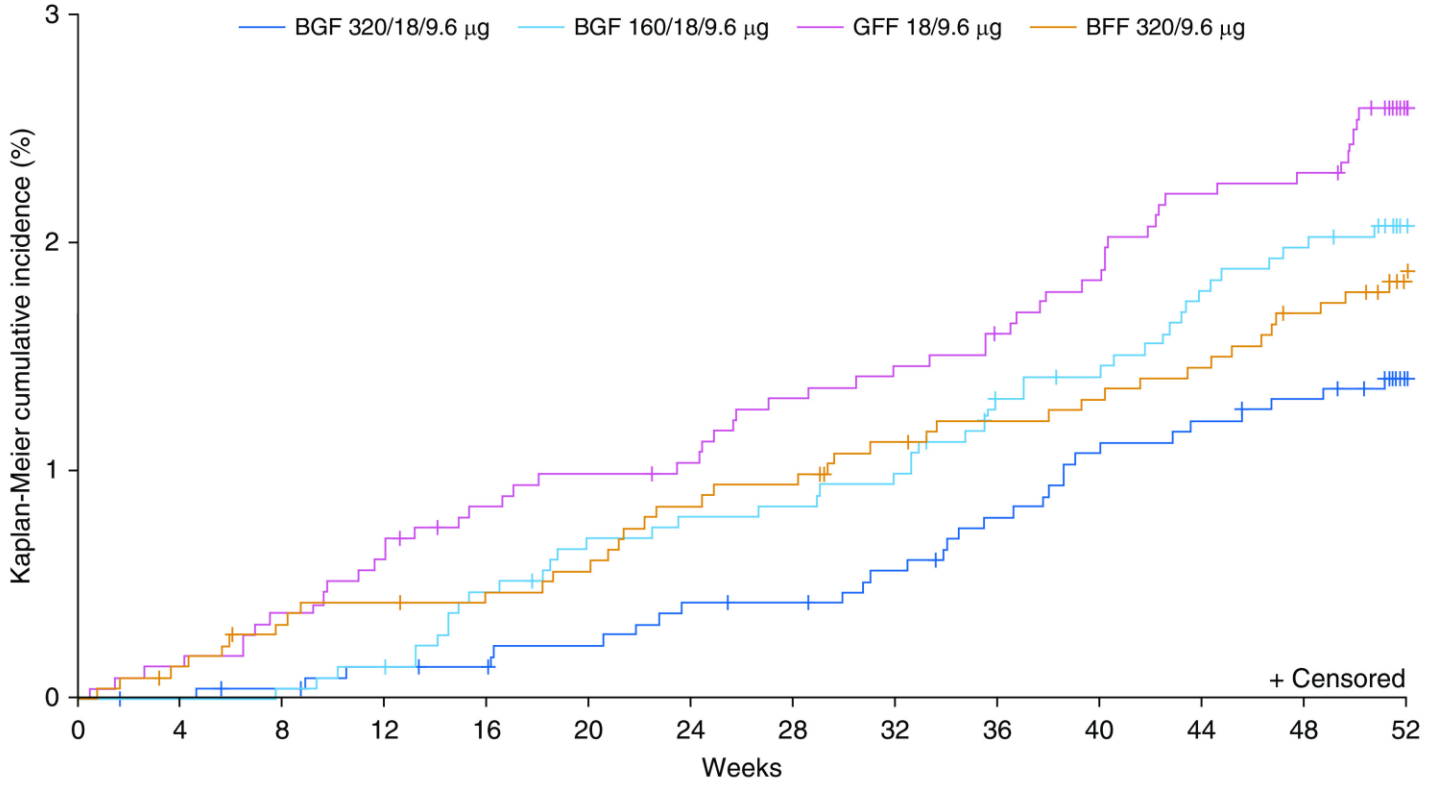


Patients at Risk

Any exacerbation					
Indacaterol–glycopyrronium group	1675	763	535	409	281
Salmeterol–fluticasone group	1679	642	415	313	217
Moderate or severe exacerbation					
Indacaterol–glycopyrronium group	1675	1299	1091	948	711
Salmeterol–fluticasone group	1679	1210	975	820	608
Severe exacerbation					
Indacaterol–glycopyrronium group	1675	1530	1434	1368	1138
Salmeterol–fluticasone group	1679	1507	1389	1303	1071

Wedzicha JA et al. N Engl J Med 2016;374:2222-2234

Inhalers – triple therapy reduces exacerbations



Patients at risk														
BGF 320/18/9.6 µg	2,137	2,136	2,134	2,131	2,130	2,127	2,123	2,122	2,118	2,112	2,106	2,103	2,100	2,075
BGF 160/18/9.6 µg	2,121	2,121	2,120	2,118	2,110	2,104	2,102	2,101	2,098	2,087	2,084	2,076	2,072	2,062
GFF 18/9.6 µg	2,120	2,117	2,112	2,106	2,100	2,097	2,095	2,089	2,086	2,082	2,077	2,069	2,067	2,045
BFF 320/9.6 µg	2,131	2,127	2,122	2,120	2,118	2,116	2,110	2,108	2,102	2,099	2,097	2,094	2,088	2,075

Triple therapy

Types of inhalers for COPD

LAMA





SABA – reliever inhalers





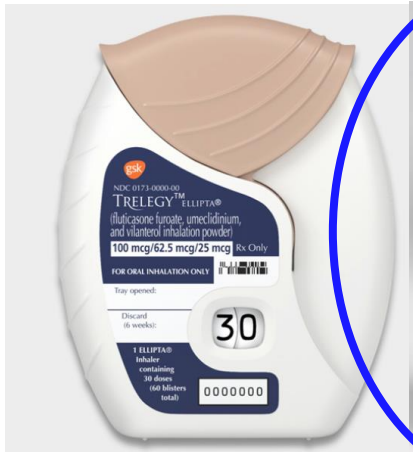
LABA/LAMA







LABA/LAMA/ICS









LABA + LAMA

Inhalers	Device Resistance	No. of steps	Evidence	Hepatic and Renal Impairment	Aerosal/ DPI	Dosage	Cost
 **Duaklir-Genuair Aclidinium/Fomoterol**	Medium	4	vs Placebo, monotherapies Dyspnoea, FEV1, QOL, Safetly profile similar	Renal-✔ Hepatic-✔	Dry Powder	Twice daily	32.50/ month
 **Ultibro- Breezhaler Glycopyrronium/ Indacaterol**	Low	10	vs Placebo, Monotherapies, tiotropium and SFC Dyspnoea, FEV1, QOL, exacerbation Safety profile similar	ESRF use only if the benefit outweighs the risk. Severe Hepatic failure - no Data	Dry powder	Once daily	32.50/ month
 **Anoro- Ellipta Umeclidinium/ Vilanterol**	Medium low	3	VS placebo, mono therapies and tiotropium Dyspnoea, FEV1, QOL, Safety profile similar	Renal- ✔ Severe Hepatic failure- no Data	Dry powder	Once daily	32.50/ month
 **Spiolto-Respimat Tiotropium/Olodaterol**		4	VS placebo, mono therapies Dyspnoea, FEV1, QOL, Safety profile similar	Caution with EGRF<50 Severe Hepatic failure- no Data	Aerosal	Once Daily	32.50/ month

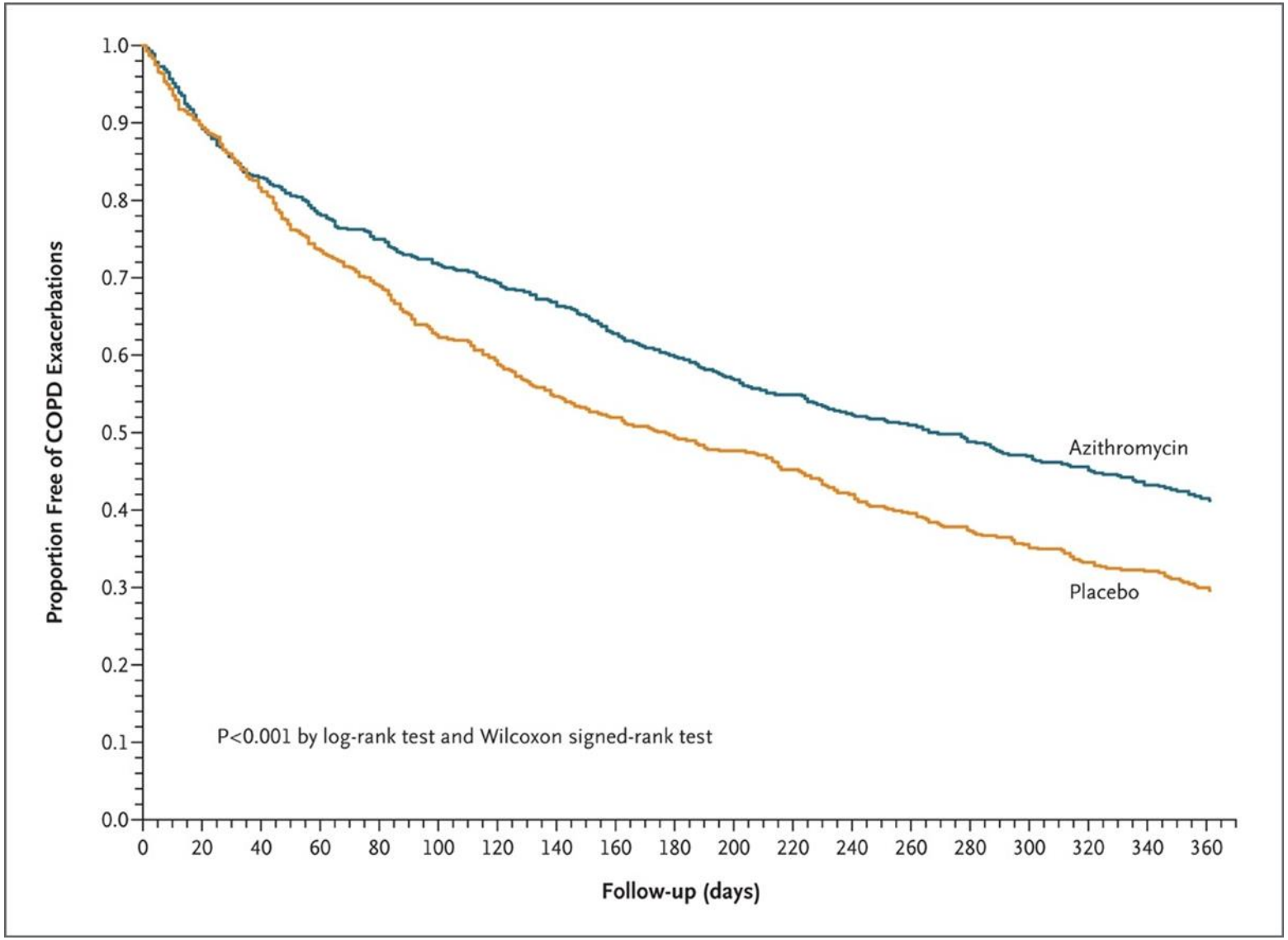
LABA+ICS+LAMA

Inhalers	Medication	Device resistance	No of steps	Steroid dose	Aerosol/DPI	Frequency	Cost
	**Trelegy Ellipta-** Fluticasone furoate/ Umeclidinium /Vilanterol	Medium Low	3	Medium	DPI	Once daily	44.50
	**Trimbow NEXT haler** beclometasone, formoterol and glycopyrronium	Medium high	3	Medium	DPI	Twice daily	44.50
	**Trimbow** beclometasone, formoterol and glycopyrronium	zero	3	medium	MDI	Twice daily	44.50
	**Trixeo Aerosphere-** formoterol, glycopyrronium and budesonide	zero	3	Medium	MDI	Twice daily	44.50

My patient still has exacerbations

What now?

Long term macrolide therapy



Check for AFB!

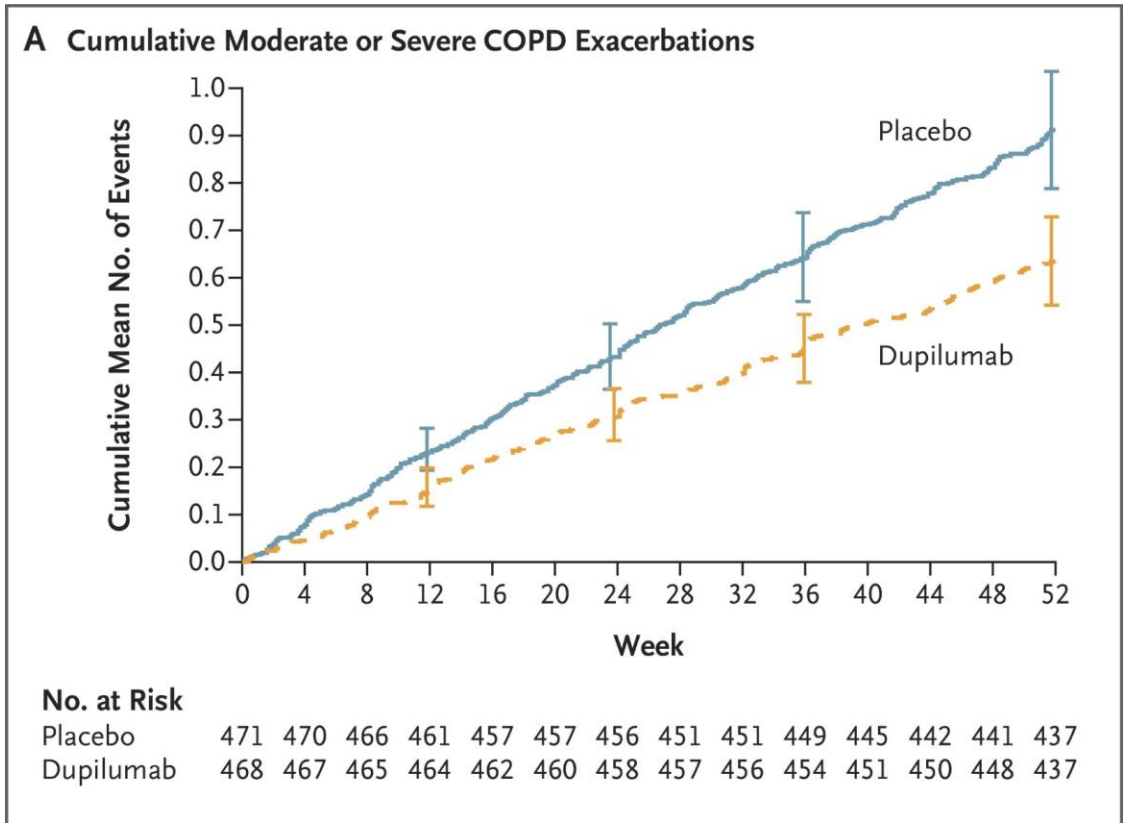
Hearing loss/ tinitus

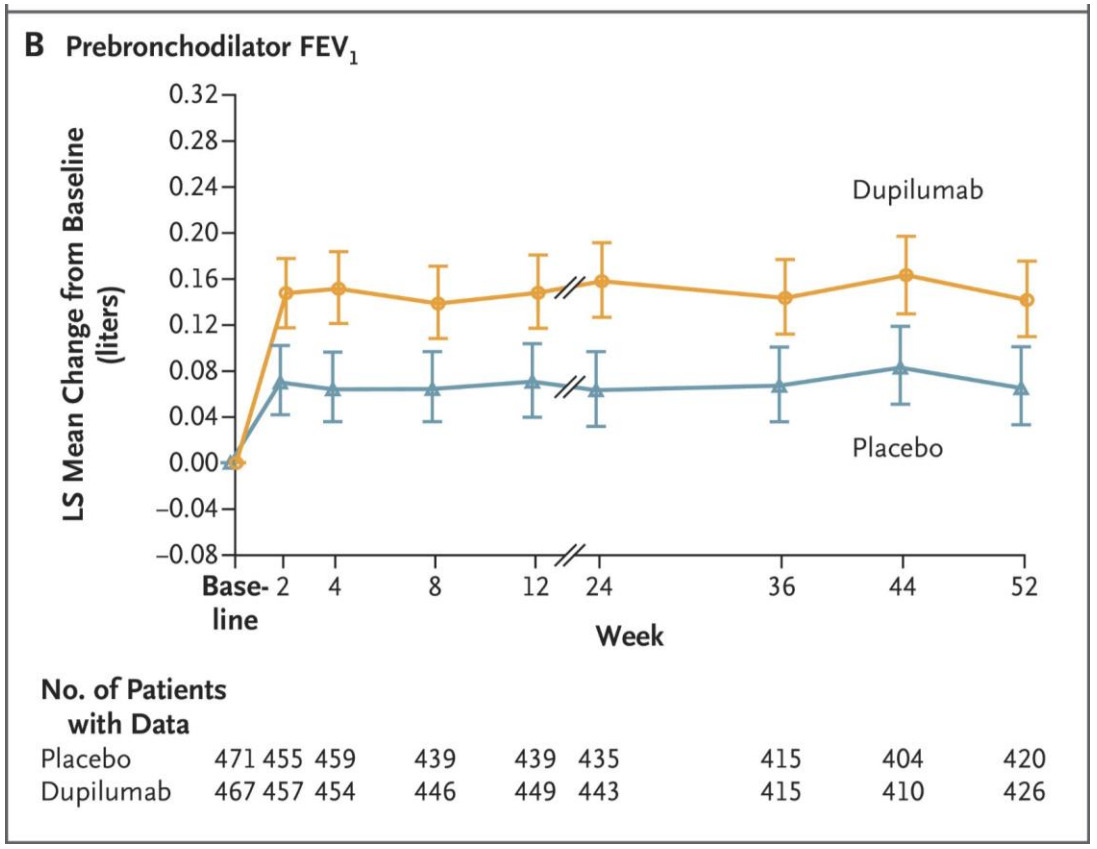
Cardiac side effects

Albert *et al* NEJM 2011

COPD Biologics

Patients with a blood eosinophil count ≥ 300 cells/μL





Bhatt et al NEJM 2023

Summary

COPD is a common and heterogeneous disease

Smoking, indoor and outdoor air pollution are common risk factors

Spirometry is used to diagnose air flow obstruction

Smoking cessation, pulmonary rehabilitation and vaccination are key to management

Exacerbations are important events and increase risk of death and disease progression

IMPERIAL

Thank you

COPD Diagnosis and Management
4 May 2026